



Think *for* Yourself

A PRACTICAL GUIDE TO
OVERCOME BIASES, AVOID DECEPTION,
AND CULTIVATE CRITICAL THINKING

THIBAUT MEURISSE

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CONTENTS

Introduction

Prologue: The impossibility of finding truth

I. Understanding our inner psychology.

1. Desire to belong
2. Desire to be right
3. Desire to be special
4. Need for congruence

II. Separating facts from feelings

1. Assessing the “factualness” of data
2. Assessing the reliability of various sources of information
3. Differentiating facts from interpretations
4. Avoiding knee-jerk thinking

III. Looking at data objectively.

1. Avoiding information overload
2. Escaping echo chambers and challenging confirmation bias
3. Protecting ourselves against ideologies
4. Avoiding being misled by statistics
5. Thinking in terms of probabilities

IV. Protecting ourselves against manipulation

1. Protecting ourselves against false experts
2. Understanding the mechanisms used by conspiracy theorists and gurus
3. Evaluating conspiracy theories
4. Exploring MLMs
5. Various scams and why we fall for them

V. Identifying patterns

1. Continuity bias
2. Exponential growth
3. Looking at trends and predictions

Conclusion

Other Books By The Authors

About the Author

INTRODUCTION

Joseph Stalin was an avid reader with a personal library of around 20,000 books. Adolf Hitler read voraciously, sometimes finishing one book in a day. And Mao Zedong was surrounded by books, often reading late into the night.

Now, the question is, were they independent thinkers? That is, were they reading to expand their worldview and improve their understanding of the world? Or were they merely looking for more “proof” that their ideology was the right one?

We may have something in common with them. At some point in our life, we may have abandoned our ability to think critically to instead embrace flawed ideologies, believe in far-fetched conspiracies, or fall for scams. For some of us, it may have been temporary and inconsequential, for others it may have lasted for years, destroying our relationships and ruining our life.

Critical thinking isn't simply a “nice to have” skill. It's not merely the icing on the top of the cake of a good life. Critical thinking is essential if we wish to avoid the many traps that we will encounter throughout our life. Each year, millions of people waste time and energy on ventures they should never have joined. Tempted by the promise of riches, these individuals jump on business “opportunities” only to end up in cult-like movements. Hoping to escape the “Matrix”, they travel down the rabbit hole of far-fetched conspiracies. Bored, they keep scrolling, slowly being seduced by extreme ideologies.

Now, what makes us abandon critical thinking? Why do we allow others to manipulate us and hijack our ability to think? Why do we get trapped in a certain ideology?

This is what we'll explore in this book.

In **Part I. Understanding our inner psychology**, we'll discuss why we have an inherent desire to belong, to be right, and to feel special, and how it can hijack our critical thinking. We'll also determine why we seek congruence and how it impacts our ability to think. Understanding our own psychology will enable us to avoid many of the traps that are discussed in this book.

In **Part II. Separating facts from feelings**, we'll determine what a "fact" is. We'll show how we're quick to dismiss facts that don't make us feel good, or how a lie told many times can often be perceived as the truth. We'll also discuss how facts can be misinterpreted and what we can do to correct this error.

In **Part III. Looking at data objectively**, we'll cover the main obstacles we face as we consume information each day. We'll discuss the danger of information overload, and we'll explain how algorithms lead us to live in an echo chamber. In addition, we'll review issues such as the misuse of statistics and other ways people distort data to manipulate us.

In **Part IV. Protecting ourselves against manipulation**, we'll see how cult leaders, gurus, and other con artists can exploit our biases. We'll unveil the psychological techniques they use to recruit us, make us believe in their false promises, or encourage us to embrace their conspiracy theories.

Finally, in **Part V. Identifying patterns**, we'll look at historical patterns that can enable us to understand reality better, think more logically, and make smarter decisions.

Developing critical thinking is not easy. Yet, when we fail to think critically, we risk travelling down the wrong rabbit hole. There are many reasons or events that can prevent us from thinking critically. For instance, it could be the loss of a loved one, mental health issues, naivety, or lack of experience. This book will shed light on the main obstacles that prevent you from thinking critically and it will give you tools to overcome them. Whenever relevant, I've added practical exercises. I encourage you to complete them.

Happy reading.

PROLOGUE: THE IMPOSSIBILITY OF FINDING TRUTH

Truth isn't something "out there", accessible to anyone who seeks it. In many ways, truth is inaccessible. In our attempt to think rationally and discover the truth, we'll come across countless roadblocks. Why is it so hard to find truth? Here are some possible answers:

- **Everyone has biases and blind spots.** It doesn't matter how hard we try to notice our biases and eradicate them, every single one of us is biased to some degree. People who think they have no agenda or consider themselves neutral, are not.
- **Human biases make the truth unknowable.** People have their own ideas of how society should be, how the economy should work, or what people should eat. The point is, we aren't the only biased ones—everybody is. As we seek to find the truth on any topic, we'll discover well-intentioned people defending their version of the truth. For instance, an MD may affirm one thing about cholesterol, while another may say something different. Or geopolitical experts may hold different theories about who is responsible for starting any given war.
- **People have hidden agendas.** It gets worse when we realize that not everyone is well-intentioned. Some people have evil intent. Those people will distort the truth to their own ends. The more technical and obscure a topic is, the easier it will be for them to do so.
- **We can't fact-check everything.** These days, we are exposed to far more information than we can digest. This negatively affects our ability to think rationally and sort out what's true from what isn't. Unfortunately, we are unable to fact-check every piece of information we encounter.
- **Truth is often unattainable.** Truth is complex and often unknowable. What we call a "fact", or an "event", is usually the result of many different issues. We often can't know all of them, nor grasp how they come together to create that fact or lead to that event. This goes for wars, economic crises, or historical events. And sometimes, we simply don't have ways to access the truth. For example, how would you prove that God exists? Truth is complex. And when we add our personal biases, ideologies,

and hidden agendas, finding the “truth” becomes an almost impossible task.

These factors put together can explain why we seem to live in a world of alternative “facts” and misinformation. But let’s remember that there can be only one fact. For instance, on February 18, 2025, Donald Trump stated that Volodymyr Zelenskyy’s approval rating was four percent. But, according to the Kyiv International Institute of Sociology, it was 57%^{*}. Both of these numbers can’t be true at the same time.

People tend to live in an alternative world inside their heads. These individuals may buy into a certain narrative and interpret events or facts based upon that narrative. While there may be dozens of competing narratives, in the real world there can only be one truth. Please bear this in mind.

^{*} Kiev International Institute of Sociology. “Public Opinion Survey: February 2025.” Kyiv: KIIS, 2025.

PART I

UNDERSTANDING OUR INNER PSYCHOLOGY

We aren't perfectly rational creatures. In many aspects, we're emotional creatures. Rather than seeing reality as it is, we perceive it through our own tinted lenses—past experiences, biological factors, emotional state, beliefs, and so on. This is what makes thinking so difficult.

In this part, we'll go over the main internal obstacles that prevent us from thinking for ourselves. These are our:

- Desire to belong.
- Desire to be right.
- Desire to be special.
- Need for congruence.

These four components encompass most of the internal obstacles that we face when we strive to make rational decisions. By understanding them, we will be able to identify our blind spots and perceive reality more objectively.

DESIRE TO BELONG

As human beings, we have an innate need to belong to a tribe. That is, we want to be accepted—to feel as though we are part of a community. In the past, we lived in small tribes and being rejected often meant dying. If we were expelled from the group, we would be left without resources, unable to feed and protect ourselves. Nowadays, we can spend most of our time alone without fearing for our lives. We can still feed ourselves, have a shelter, and live a relatively normal life. In other words, being accepted isn't essential for our survival. And if we are rejected from one group, we can always find another.

Yet, because of our innate need to fit in, we often act as if other people's approval were the most important thing in the world. As a result, we would rather agree with others than speak out in opposition. In addition, following the crowd gives us a sense of security. We can feel that if everybody else is doing it, they must be right.

Wall Street analysts demonstrate a good example of herd mentality. Many analysts chase stock prices. That is, rather than making predictions based on independent thinking, they wait for a company stock price to surge before increasing their price target. This isn't thinking, it's reacting.

This pattern is obvious when you look at stocks of companies such as Tesla. Tesla stock surged from \$250 to \$480 following the election of Donald Trump as President of the United States. Suddenly, most analysts raised their forecast to \$400 or \$500. Yet, there were no fundamental changes in Tesla as a company. No new products were announced, and Tesla still had the same company culture, employees, CEO, and roadmap as before. Now, they may have seen Donald Trump's support of Elon Musk as a catalyst, but did it justify a sudden raise of their target?

Truth is, for Wall Street analysts, like for most people, there is security in numbers. By choosing a target similar to the current price of the stock, they don't have to stand out. They don't have to risk being the only ones to be wrong, which could damage their reputation.

This herd mentality manifests itself in many other ways. For example, managers usually don't get fired for shutting down great ideas. In many companies, the employees who earn a promotion are the ones who avoid rocking the boat. In other words, they have a bigger incentive to maintain the status quo than to try something new. This is one of the reasons companies can miss new technologies and go bankrupt. Entrenched in their current ways of doing things, they're unable to pivot and adopt new technologies. A famous example is Kodak. They invented the digital camera but failed to capitalize on it. Or think of Blockbuster Video. They had thousands of physical stores but disappeared when they failed to embrace video streaming.

Authority bias

Thinking for ourselves requires humility, curiosity, and a desire to learn from others. However, it also demands that we stand up for what we believe in, and challenge others even when it feels uncomfortable. But this is hard, which is why, instead of thinking for ourselves, we often outsource our thinking to so-called experts. We think, "They're smarter than us, so they must know better." But we should never abdicate our critical thinking. Human beings' reluctance to challenge authority has been responsible, at least partially, for some of the most atrocious crimes in history.

The three experiments below illustrate how easily we can be influenced under certain conditions. They are a powerful reminder that no one is immune to a lack of critical thinking and that our need to belong can open the door to all sorts of manipulation.

The Milgram experiment

Stanley Milgram, a psychologist from Yale University, wanted to understand why the Holocaust happened. He asked whether it was possible that people involved in the Holocaust were just following orders. And how could that be? So, he designed an experiment where one participant (the teacher) would punish another participant played by an actor (the learner) by delivering an electric shock when the learner gave an incorrect answer. The learners were hidden from the teachers, but the teachers could hear the learners screaming. During the experiment, the teachers were unaware that no electric shocks were actually being delivered. The screams were recorded. The "doctor"—an actor wearing a white lab coat—told the participants to continue even when the learner was screaming and begging the teacher to stop.

Towards the end of the experiment, the learner would scream:

"Experimenter! That's enough. Get me out of here! I told you I have heart trouble. Get me out of here, please!"

So, what were the results?

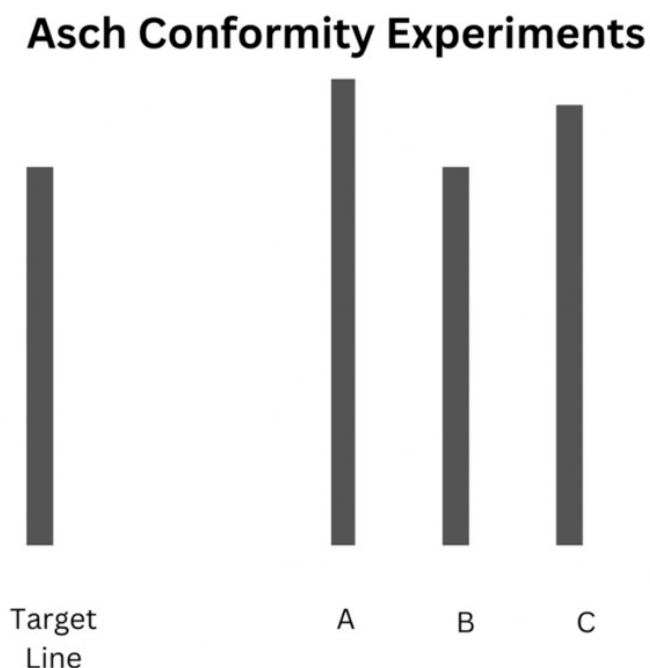
Before the experiment, Milgram polled fourteen psychology majors as well as some of his colleagues. They believed that only a small fraction of the participants would

be willing to inflict the maximum charge (450 volts). However, the experiment revealed that 65% (26 out of 40 participants) did so even though the learner was screaming, begging them to stop.

This experiment showed that authority figures can lead us to become obedient and do things that we would never have done otherwise. The bottom line is, no matter how strong we think we are, overcoming our tendency to comply with the instructions of figures in authority is hard.

The Asch conformity experiment

In this 1950s experiment, the psychologist Solomon Asch wanted to test how group pressure influences our actions. Participants were shown a card with four lines on it, and they were asked to indicate which one matched a separate target line. In each case, the correct answer was obvious. But the trick was that only one of the participants in the group was real. All the other participants were actors. The results were surprising in that 75% of the real participants agreed with the false participants (the actors) at least once, when the answer was blatantly wrong. This experiment emphasizes our innate desire to conform. It's another reminder that thinking for ourselves is hard and should never be taken for granted. Below is an example of a card shown to participants.



The Stanford prison experiment

To study how people react to a negative environment, psychologist Philip Zimbardo turned Stanford's psychology building into a prison. He wanted to know whether bad situations could turn good people into bad people or if our attitudes, our values, and our morality could act as a protective shield.

Twenty-four students were hired to play the role of the guards and the prisoners. Although the experiment was supposed to last two weeks, things deteriorated so quickly that it had to be interrupted after only six days. Just 36 hours in, one prisoner had already started to act illogically.

One third of the guards engaged in sadistic behavior. They forced some prisoners to defecate in a bucket placed in their cells and ordered them to repeat their assigned numbers again and again. They also practiced physical punishments. Several “prisoners” were traumatized and had to be released. Philip Zimbardo himself, who was acting as a superintendent, admitted later that he was so caught up in his role that he didn’t realize the gravity of the situation.

These three notorious experiments show how challenging it can be to go against the grain. People appear hardwired to seek consensus and follow what everybody else is doing. It seems counterintuitive to stand for what we believe when we face peer pressure or receive orders from perceived authority figures. In many cases, individuals would rather feel accepted and belong than be rejected. As a result, our default answer is often to mimic other people’s behavior.

What about you? How would you have reacted if you had participated in these experiments? Would you have been able to protect your ability to think for yourself? Or, like most people, would you have succumbed to the desire to conform?

Ultimately, thinking for ourselves requires making decisions because we believe they are the right thing to do. The more we practice doing this, the easier it will become. But we must remember that there will always be a strong desire for us to fit in.

* * *

Action step

Complete the following exercises in your action guide:

- Give three past or current examples that show you’re trying to belong rather than thinking for yourself.
- Come up with a couple of examples that show others around you are trying to belong.
- Reflect on the following: Have I ever blindly followed an authority figure when I should have thought for myself? Have I ever conformed due to social pressure when I had a different opinion or wanted to act differently? Have I ever lost myself in some kind of role, only realizing it went against my better judgment afterward?

DESIRE TO BE RIGHT

As humans, we also have the desire to be right. As we go through life, we build our unique model of reality based on our experiences, upbringing, and set of beliefs. In other words, we create a certain sense of identity. Then, we spend time and energy trying to protect that identity. The problem is, it's difficult to think for ourselves if we refuse to acknowledge when we're wrong. Thinking critically requires a willingness to challenge our beliefs. And most people aren't willing to do that. These individuals would rather keep being wrong than make changes in their lives.

Everyone is right

Most of us think we are right and believe we are doing the right things. Otherwise, we would change our thinking and act differently. Even Hitler, Stalin, and Mao believed they were doing the right things. But not everybody can be right. To become independent thinkers, we must be willing to change our mind.

One of the best questions you can ask yourself is:

How could I be wrong here?

Pride vs. learning

Our desire to be right often prevents us from learning and growing. We would rather win an argument or keep going down the wrong path with our career, beliefs, or relationships than acknowledge that we are wrong.

When we insist on being right, we cling to an old identity. Often, it leads us to keep doing what we've always done before. This is not thinking. This is living on autopilot. To let go of pride, one solution is to see ourselves as learners. By choosing the identity of a learner, we can overcome our desire to always be right. Perceiving ourselves as a learner means that we seek pleasure from changing our mind. We experience joy from learning new things and from exposing ourselves to new ideas. As a result, we will be willing to abandon some of our misguided pride in the pursuit of truth and learning.

When we see ourselves as learners, our identity becomes closely linked to our ability to update our worldview. Instead of resisting change, we will embrace it. Change becomes a central component of who we are.

To conclude, we want to be right, which can lead to misguided pride and prevent us from learning. The solution is to see ourselves as continuous learners.

Confirmation bias

We generally don't seek information that challenges our worldview. Instead, we look for information that confirms what we already believe. Questioning our beliefs would force us to admit that we may have been wrong. It would affect our pride and self-esteem. We would rather avoid doing that.

For example, someone who identifies as a conservative will only watch the news from right-wing media such as Fox News or the Daily Wire. Likewise, someone who identifies as a liberal will only watch liberal media, such as CNN or MSNBC. As a result, it's unlikely that they will change their minds or consider opinions from the other side as having any merit.

Conservatives and liberals sometimes distort the facts to fit their narrative. This is why we must exert critical thinking to differentiate between facts (what actually happened) and fiction (erroneous narratives).

Refusing to believe people we hate

When we hate a public figure, we create a narrative around them. We make assumptions regarding the way they think, their intentions, and so on. In short, we believe they are wrong while we're right (or more right than they are). As a result, we dismiss everything they say, including actual facts. While this is understandable, it is the opposite of thinking. To become independent thinkers, we must separate our feelings from the facts. We don't have to agree with what someone says, but we should look at their argument objectively and give them credit when it is deserved. Otherwise, we'll end up in an echo chamber.

For instance, we may hate politicians, whether Donald Trump or Joe Biden. Yet, some of their ideas might be good. Or we may despise Elon Musk or Jeff Bezos, but we can still learn useful business lessons from them.

The bottom line is, we must be aware of our hatred of public figures. It's often a sign that we have become ideologues and have abandoned our ability to think for ourselves. Similarly, we must notice when we find ourselves agreeing with everything someone says. In which case, we might have become too naive or too lazy with our thinking.

Excuses and justifications

Sometimes, rather than trying to change our situation, we blame our circumstances. While circumstances do impact our lives, we must have enough self-awareness to

see what we could do about it. In other words, thinking for ourselves requires us to avoid making excuses. Below are some examples:

- **I'm out of shape because of the food industry.** Being fit and healthy isn't easy and requires consistency and planning. But if we want to get in shape, the first step is to acknowledge that our current situation is the result of our behavior and the choices we made.
- **I can't save because of inflation.** It can be hard to save money, but we sometimes live above our means. For example, people may buy a new car or rent a place that is more than they can afford. In doing so, they have no margin of safety if something unexpected happens. It is our responsibility to acknowledge that the future is unpredictable and prepare accordingly. This entails living below our means, avoiding debt, and saving as much as we can.

Whenever you notice that you're making excuses, ask yourself, "Is that really true or am I lying to myself?"

* * *

Action step

Answer the following questions using your action guide:

- How are you trying to be right? Give two to three examples from your own life.
- In what ways are you seeking to confirm what you already believe?
- If you only cared about learning and building an accurate perception of the world, what would you do differently?
- What excuses are you making to escape the harsh truth that you must make changes?

DESIRE TO BE SPECIAL

While we may not like to admit it, part of us wants to feel superior. We want to believe that we are better than others and know things that they don't. This need to feel special is not necessarily a bad thing, but it can lead us astray by numbing our critical thinking.

On the positive side, our desire to feel special may lead us to cultivate interesting skills, enhance our self-esteem, and bring joy and meaning into our lives. For instance, I want to become an excellent writer and sell millions of books because it makes me feel good. It makes me feel special to be able to achieve a goal that most people can't or won't manage. For others, that desire to feel special may manifest in different ways. They may aspire to learn several languages, become a world-class chess player, or know more about basketball than anyone else.

The bottom line is, embracing our desire to be special can be an excellent way to acquire more skills and make the most of our talents. After all, we all want to be proud of who we are and what we do. However, on the negative side, our desire to be special can also become an issue. For example, it can lead us to embrace extreme ideologies or believe in far-fetched theories.

Why we believe in conspiracy theories

Our desire to be special is one of the reasons we can end up believing in conspiracy theories. In truth, many people believe in conspiracy theories. But why is that?

The world is complex and nuanced. As a result, we must rely on simplified models in an attempt to understand it. Conspiracy theories tend to be oversimplified models of reality. They create the illusion of certainty and predictability in an uncertain and unpredictable world. They give us the impression that we understand how the world works. In addition, they make us feel as though we know some secrets most people ignore. Finally, they enable us to channel our frustrations and anger toward a common enemy.

In other words, conspiracy theories don't merely tap into our desire to be special, but they also key into our desire to be right and to belong. They make us believe that we know the truth (desire to be right), and that we must stick together to fight a common enemy—illuminati, global elite, international bankers, et cetera (the desire to belong). This is what makes these theories so enticing. The same mechanism is at play with cults or extreme ideologies. Later in this book, we'll talk about the psychological manipulations used by these groups to recruit people and erode their ability to think.

In conclusion, our desire to feel superior and good about ourselves can prevent us from thinking critically. It might lead us to believe in conspiracy theories or to embrace extreme ideologies. This is why we must stay vigilant and keep our desire to be special under control.

* * *

Action step

Answer the following questions in your action guide:

- How are you trying to be special? Is it making your life better or worse?
- What conspiracy theories do you believe in, or did you believe in previously? Why do you think you fell for them? Do you still think they are true?
- What unconventional beliefs do you hold? What would you say is the likelihood that they are correct?
- Now, how could you use your desire to feel special in a more constructive way?

NEED FOR CONGRUENCE

We also have a need for congruence—that is, we want the actions we take to be aligned with our identity (the way we see ourselves). Let's see how the desire for congruence can lead us to fall for biases. Then, let's discuss what we can do to protect ourselves against it.

Cognitive dissonance

Cognitive dissonance is the mental unease we experience as a result of holding two or more conflicting beliefs, values, or attitudes at the same time, or from acting in ways that go against our beliefs or values. Our need for congruence leads us to use any justification possible to resolve these inner contradictions. Below are some examples of cognitive dissonance:

- A self-help guru may tell his audience that they should strive to add value to people's lives while running a multi-level marketing company (MLM) where most members lose money. He may resolve the dissonance by lying to himself pretending he's offering people genuine opportunities.
- Someone, attracted by the promise of a high salary, accepts a job working for a tobacco company even though they hate tobacco. They resolve the issue by rationalizing, "If I don't do that job, someone else will", "Who am I to tell people whether or not they should smoke?", "New tobacco products are less harmful", and so on.
- A person identifies with a political party that claims to value integrity, but they learn that their favorite politician has lied. They resolve the dissonance by minimizing the lie or accusing the media of being biased.

We all experience cognitive dissonance in some areas of our lives. To think for ourselves, we must look in the mirror and address it wherever it may appear.

Rationalization

We like to think of ourselves as being rational. That's why we often strive to find reasons to justify our actions after the fact. This is referred to as "rationalization."

For example, you may hate your job but tell yourself it's not that bad because it pays well. You may go to a fancy restaurant and convince yourself that the food was good just because it was expensive. Or you may eat an entire pint of ice cream and argue that you deserve it because you had a stressful day at work.

We all rationalize actions after the fact. Thinking more critically requires that we become aware of rationalization and try to do it less often.

Sunk cost fallacy

Often, we keep investing our time, money, or energy into a venture just because we don't want our previous investments to go to waste. Ironically, this leads us to waste even more resources. This is an extreme example of rationalization. We've been wrong but prefer to continue along that path rather than changing course. This is referred to as the "sunk cost fallacy."

In truth, because we've invested a lot of resources into an endeavor doesn't mean we should keep doing so. As the saying goes, doing something that doesn't work over and over is a sure sign of insanity. When we fall for the sunk cost fallacy, we abandon critical thinking and waste our precious resources. We don't need to keep doing what doesn't work or what we don't like. We can pivot and reinvest our time, energy, and money into something else we deem more fruitful. For instance:

- Instead of staying in an unsatisfying relationship, we can leave and eventually find a better partner.
- Instead of staying at a soul-crushing job, we can look for a more satisfying career.
- Instead of investing more time and money into a failing business venture, we can do something else with a higher chance of success.

The foot-in-the-door technique

Once you've started travelling down a certain path, you'll tend to keep going. Marketers know that and try to exploit this bias. This is often referred to as the "foot-in-the-door technique."

My girlfriend's mom was offered a free beauty consultation. She was excited about it and reserved a spot. But my girlfriend convinced her not to go. She understood that, once her mom was at the salon (one foot in the door), they would seize the opportunity to sell her something she probably didn't need or want. On a side note, a technique often used by unscrupulous marketers is to tell their prey they have a problem and must do something about it. If my girlfriend's mom had shown up to the consultation, they would likely have found serious issues with her skin and offered her the perfect (and expensive) "solution".

Here is another example. Years ago, my girlfriend visited the dentist to solve a small issue, but the dentist suggested she put veneers on all her teeth, which

would have cost tens of thousands of dollars. She had a foot in the door, but it was way too expensive, and, fortunately, she didn't go along with his suggestion.

The key takeaway is, because we have a foot in the door doesn't mean we should put our entire body in. We can just remove our foot and leave.

The commitment bias

Once we've made a decision, we tend to keep going because of our desire for congruence. Commitment bias is another example of that. Once we commit to something, it's often hard to back off.

During the Korean War, Chinese communists attempted to get American prisoners of war (PoWs) to change their political views. They used their knowledge of the commitment bias to do so. For example, PoWs were asked to make minor concessions, such as writing down slightly critical comments about the US government (e.g., "America isn't perfect"). Later on, they were asked to elaborate and write short essays detailing why capitalism isn't perfect. After that, they were asked to read it out and post it somewhere where people could see it. This increased their sense of commitment and strengthened their need to stay congruent.

Unconsciously, PoWs may have felt, "Why would I assert this publicly unless I believe it?" It created cognitive dissonance that needed to be resolved. Over time, they were encouraged to make bigger commitments, such as participating in group discussions, attending lectures, or even making radio broadcasts. As a result, some PoWs ended up championing communist views after their release.

Emotional reasoning

We often believe that what we feel must determine what we do—i.e., there should be congruence between our feelings and our actions. This is called "emotional reasoning." But we don't have to feel like doing something to do it. Otherwise, we wouldn't need self-discipline. What makes us human is that we can choose. We can choose to do something difficult or unpleasant now because it will benefit us long term. For example, we may not feel like working, studying, or applying for a new job, yet we still do it because it will benefit us in the long run.

What about you? How often do you fail to do what you need to do because you don't feel like it (e.g., you're tired, scared, or unmotivated)?

To sum up, we tend to feel the need to be congruent with our feelings of the moment. But what determines a major part of our success is what we choose to do when we *don't* feel like it. Thinking critically means understanding that we shouldn't let our feelings determine our actions, or at least not all the time.

In this section, we've seen that we tend to abandon critical thinking because of our desire to belong, to be right, to feel special, and to remain congruent. In the next section, we will see how our feelings can interfere with the facts and prevent us from thinking critically.

PART II

SEPARATING FACTS FROM FEELINGS

Facts don't care about our feelings.

That is, what is true is true, whether we like it or not. To think more critically, we must be willing to face the truth and accept it. In this section, we shall go over ways to separate facts from feelings so that we can sharpen our critical thinking skills.

ASSESSING THE “FACTUALNESS” OF DATA

We all consume an unfathomable amount of data each day. However, each data point isn't created equal. While some data is signal (relevant), most of it is noise (irrelevant). Let's first explain the different types of information we're exposed to each day and see how reliable they are.

The hierarchy of “facts”

It can be useful to think of data points as part of a hierarchy. That is, some types of information are fairly accurate (close to the truth) while others are merely anecdotal (unreliable). To improve our thinking, we must sort out the information we consume—i.e., we must differentiate “signal” from “noise”. Here is the hierarchy of facts we'll discuss:

1. Laws of physics.
2. Verifiable facts.
3. Observations/Personal experiences.
4. Assumptions/hypotheses.
5. Logical reasoning.

A. Laws of physics

It's hard to deny the laws of physics because they are observable. For instance, everybody can see that gravity is real. We also know the speed of light or the physical properties of certain materials. In short, we understand when the laws of physics apply and when they don't, and we can measure them.

B. Verifiable facts

A fact is something that we know to be true beyond any doubt and data we can verify. For example, few people would argue against the fact that the sun rises every day. We can verify that each morning. Below are some verifiable facts:

- The Declaration of Independence was signed in 1776.

- Fish live in water.
- Humans need to eat to survive.
- Mount Everest is the tallest mountain above sea level in the world.
- The iPhone was first released on June 29, 2007.

C. Observations/personal experiences

Our ability to see with our own eyes is important. Many so-called facts can be manipulated. For instance, journalists may distort facts to make the story conform to their political affiliations. And people are quick to share any news that fits their worldview without doing proper fact-checking. That's why, as biased as we can be, seeing things with our own eyes is still one of the best ways to know whether something is true or not. I can believe that the moon landing is a hoax, no matter how many pieces of evidence I'm presented with, but if I were Neil Armstrong, I would know it for a fact.

The point is, it's healthy to question everything that we haven't seen ourselves. However, at the same time, we must avoid unwarranted skepticism. Otherwise, we risk believing in all kinds of far-fetched theories and other fake news.

Words of caution: we can all be prone to hallucinations. Because Joan of Arc thought she heard voices from saints doesn't mean it was real. And because we think we saw God doesn't mean we did. We all want to feel special, but just because we're convinced of something doesn't mean it actually happened.

D. Assumptions/hypotheses

Assumptions are what we think could be true based on our personal experiences and the information we have. An assumption isn't a fact. It could be accurate, but it could also be wrong. To think critically, we must differentiate between assumptions and facts.

Below are examples of assumptions:

- John didn't say "Hi" this morning at work, so he must be angry with me.
- I will beat my friend at tennis this coming Saturday.
- There will be a recession this year.

These assumptions could turn out to be true or false. Perhaps John didn't say "Hi" because he was preoccupied with serious issues. Perhaps my friend will actually beat me at tennis. And perhaps there will be no recession this year.

To know whether something is an assumption, let's cultivate the habit of asking ourselves if it's actually true:

- Am I *absolutely* sure that John didn't say "Hi" because he is angry with me?

- Am I *absolutely* sure that I will beat my friend at tennis?
- Am I *absolutely* sure that there will be a recession this year?

In doing so, we'll realize that facts aren't all that common, whereas assumptions are plentiful.

People sometimes talk with so much confidence that everything they say seems to be the truth. But usually, what they present as facts are merely assumptions, opinions, or hypotheses. Please keep this in mind.

E. Logical reasoning

We must be able to infer logical connections between various events or pieces of information to make sound decisions. However, it's easy to use what we believe to be logic and still arrive at the wrong conclusions. This is what many conspiracy theorists do when they use "logical reasoning" to "connect the dots." Thus, while making logical connections can be useful, it doesn't necessarily mean we're getting closer to the truth. To ensure your reasoning is correct:

- **Check your premises.** If your assumptions are wrong, everything that follows will be wrong, too. For example, if your premise is that a group of evil people control the world, then you'll see many of the world issues as part of their Machiavellian agenda.
- **Zoom out.** Things are often part of a complex system. If you consider something in isolation, you risk missing the bigger picture. For instance, health influencers sometimes isolate a specific compound to prove a food item is unhealthy. Paul Saladino, an MD proponent of the carnivore diet, claims that broccoli is dangerous because it contains isothiocyanates—a compound that can hinder the absorption of iodine by the thyroid gland. However, except for a minority of people with certain underlying conditions, studies show positive benefits from eating broccoli.
- **Consider the second-order consequences.** Our actions can create a chain of reactions. For example, a company may lower their price to get more customers (first-order consequence) and by doing so, start a price war (second-order consequence). As a result, they may end up going bankrupt due to lack of profit (third-order consequence). We must be aware of the unintended consequences our actions may lead to.

Below are specific examples of "logical reasoning" that went wrong:

Working out. Working out increases heart rate, blood pressure, inflammation, and oxidative stress. Therefore, it is possible to assume that exercise must be bad for us. However, study after study has shown exercising to be beneficial to our overall long-term health. The fallacy here is to believe that what would be a problem if it were chronic (increased heart rate, higher blood pressure, inflammation, oxidative stress, et cetera) is also beneficial when temporary (when working out).

Automation. With automation, people will lose their jobs. Therefore, we should stop automation. Yes, with new technologies, millions of people do lose their jobs, but new jobs are often created as a result. Coachmen were put out of business when cars displaced horse-drawn carriages. Coal miners found themselves out of a job when machines replaced them. Cashiers lost their jobs because of self-checkout. The issue here is to see the job market as a zero-sum game. New technologies destroy jobs but they also create new jobs when people move from old industries to new ones. Consequently, the job market and the economy must be considered in their entirety.

Logical reasoning, when used well, can enable us to move closer to the truth. On the other hand, if we start with false premises, fail to understand the bigger picture, or ignore second-order consequences, we may arrive at the wrong conclusions.

ASSESSING THE RELIABILITY OF VARIOUS SOURCES OF INFORMATION

Now that we've seen the hierarchy of facts, let's go over various pieces of information and see how to assess their reliability. In this book we'll explore the following sources:

1. Articles.
2. Research papers.
3. YouTube content.
4. Books.
5. Conversations.

A. Articles

Some articles are well-researched and insightful, while others are highly ideological and manipulative. To analyze the reliability of an article, we should consider the following:

- **Facts vs. Interpretations.** What are the facts, and what is the writer's interpretation of those facts?
- **Validity of the Facts.** What are the sources? How credible are they? How do we know they are true?
- **Writer's Opinion.** What is the writer's opinion? Do they have biases because of their political leanings or their personal ideologies?
- **Expertise.** How knowledgeable is the writer on the topic? Why should we listen to them?
- **Motivations.** What are the writer's motivations? What are they trying to do? Are they trying to entertain us? Do they seek to inform us? Or is their goal to go viral?
- **Conflicts of Interest.** Are there possible conflicts of interest?

We cannot fact-check everyone all the time, but we can build the habit of separating facts from interpretations or opinions. In doing so, we will become better

at thinking independently.

Now, here is how you can fact-check articles.

Step #1. Go to the source of the information. Whenever possible, go directly to the source. Read the article cited or watch the interviews mentioned. Sometimes, so-called facts are taken out of context or they can be grossly misinterpreted.

Step #2. Cross-check using multiple fact-checking websites. There are many fact-checking websites. Visit a few of them and search for the main claims made by the writer. The table below lists what currently appear to be the most reliable fact-checking websites.

 Fact-Checkers by Political Lean

Lean	Fact-Checker	Description & Strengths
Center	Reuters Fact Check	Very high credibility, neutral tone, global focus.
	AP Fact Check	High reliability, minimal editorializing.
	FactCheck.org	University-backed (Penn), methodical and transparent.
Center-Left	PolitiFact	Poynter-run, widely cited, uses "Truth-O-Meter" system.
	Snopes	Covers a wide range of claims (politics, pop culture, etc).
Center-Right	The Dispatch Fact Check	Conservative, but credible; focuses on disinfo.
Right	CheckYourFact	Right-leaning Daily Caller spinoff; generally well-sourced.

Step #3. Ask ChatGPT, Grok, or other similar large language models (LLMs). Copy the link to the article into an LLM (large language model) like ChatGPT and ask it to fact-check the article. Then, cross-check across a number of LLMs.

These three steps should give you a better understanding of how reliable an article might be.

* * *

Action Step

Using fact-checking websites and LLMs, fact-check the following claims:

- Elon Musk's claim that President Joe Biden's immigration policy was "a giant voter importation scam."
- Raising the federal minimum wage to \$15/hour would lift millions of Americans out of poverty.
- "The Earth's average surface temperature has increased by approximately 1.1°C (2°F) since the late 19th century."

Don't forget to go to the sources when possible and to use multiple websites to cross-check.

B. Research papers

Research papers are a great way to deepen our knowledge on various topics. But their reliability varies, and many findings can't be replicated. It has been suggested that around half of published studies may not be replicable in independent tests.

Now, let's see how to assess the quality of various research papers. For the data to be as reliable as possible, we should focus on systematic reviews or meta-analyses published in reputable journals. Below is a hierarchy of study types, from most to least reliable.

Systematic reviews and meta-analyses. These studies synthesize data from multiple studies (often randomized controlled trials). They are considered by many to be the most reliable types of studies.

For instance, a 2013 meta-analysis^{*} published in *The Lancet* examined the efficacy of statins (cholesterol-lowering drugs) in reducing cardiovascular events. The study included data from 26 randomized controlled trials (RCTs) and involved over 170,000 participants. It found that statins significantly reduced the risk of major cardiovascular events.

Randomized controlled trials (RCTs). In these studies, participants are randomly assigned to a group, either the intervention group (the group receiving the treatment) or the control group (the group receiving a placebo or standard treatment). They follow a rigorous scientific method to minimize bias and establish causality.

For instance, in the Diabetes Prevention Program^{*}, over 3,000 people were randomly assigned to lifestyle changes, medication, or a placebo. The study showed that simple lifestyle changes like diet and exercise were more effective than medication in preventing type 2 diabetes.

One caveat: RCTs on animals are less reliable. What applies to animals doesn't necessarily translate to humans.

Cohort studies. These observational studies follow groups over time to identify associations.

For example, researchers might follow a group of smokers and a group of non-smokers and see how many people develop lung cancer over ten years. These studies can be useful and show correlations, but they can't definitively establish causality.

Case-control studies. These are also observational studies, but they are completed retrospectively. They can be useful in finding associations, but as with cohort studies, they can't establish definite causality.

An example is the 1950 study by Wynder and Graham, which investigated the link between smoking and lung cancer. When comparing the smoking habits of 684 men with lung cancer to those without, they found a strong association between smoking and lung cancer incidence^{*}.

Each of these studies has its pros and cons. Meta-analyses tend to be more reliable provided they have access to good data from various RCTs. RCTs also offer a high degree of reliability but can be time-consuming and hugely expensive. Cohort and case-control studies are useful for correlations but less definitive in establishing causality.

How to navigate research papers

To make the most of research papers and avoid falling for biases, consider the following:

Start with systematic reviews or meta-analyses. We can easily become lost in the sea of misinformation online and fall for dubious claims. Everyone has their own agendas. Therefore, instead of listening to your favorite influencer, see if you can find systematic reviews or meta-analyses. If not, extend your research to human randomized controlled trials. You can do so using the following websites:

- PubMed: <https://pubmed.ncbi.nlm.nih.gov/> —> Filter for “review,” “meta-analysis,” or “randomized controlled trials.”
- Google Scholar: <https://scholar.google.com/> —> Search: “meta-analysis,” “systematic review,” or “randomized controlled trial” + your topic.

Make sure you look at the most recent studies.

Check the reliability of a journal. Ideally, you want to focus your research on papers published in well-established, peer-reviewed journals. You can visit the following website to assess which journals are reliable: <https://www.scimagojr.com/>

On the page below, they explain how each journal is ranked:

https://www.scimagojr.com/help.php#understand_journals

Use aggregators. Not everybody has time to read studies. Fortunately, AI can help you summarize studies. For instance, you can use <https://scispace.com/>. Let's say you want to understand whether statins are effective at reducing the risk of

cardiovascular diseases. You can type: “Is statin helpful in reducing the risk of cardiovascular diseases?” Alternatively, you could use ChatGPT or other LLMs. Here are some examples of prompts you can use:

- “Summarize the key findings of the most cited papers on [specific medical condition] in the last five years.”
- “What are the limits of [insert specific research paper cited by your favorite influencers]?”
- “Give me a list of meta-analyses or randomized controlled trials on [insert topic of your choice].”

When you write prompts, make sure to be as neutral as possible. For instance, don’t ask, “Find me studies that show how a high level of LDL cholesterol increases the risk of cardiovascular diseases.” Instead, ask, “What is the relationship between LDL cholesterol and cardiovascular disease risk?”

* * *

Action step

Using what you’ve learned in this section, try to find answers to the questions below:

- Does a high LDL cholesterol level increase the risk of cardiovascular diseases?
- Does a vegan diet produce better health outcomes long-term—i.e., lower risk of chronic diseases and mortality?
- Which is better for long-term health, resistance training or cardiovascular training? And should you do both?

C. Books

I love books, and I believe they are the most cost-effective way to learn almost anything. However, their quality may vary greatly. Often, the authors have an agenda or fall prey to various biases due to their life experiences, political leanings, or personal beliefs. They may also lack the required expertise. When reading a book, consider the following questions:

- **What are the political leanings and main ideas the author believes in and tries to promote?** Are they left-leaning or right-leaning? What are their views on key topics like climate change, economic policies, immigration, or anything relevant to the topic they’re writing about?
- **What is the author’s expertise on the topic?** Do they hold a PhD? Do they have decades of experience on the topic? Do they have a track record

that makes them a credible source of information?

- **Is it scientifically based?** Are the author's claims based on science? If so, do the studies they refer to reflect the current state of science accurately, or is the author cherry-picking data?
- **Who supports the author?** Who is endorsing the book? Who is interviewing the author or recommending their book? Who's the audience?
- **What do reviews say (positive and negative ones)?** Does the book have good reviews? What are the negative reviews saying? What potential shortcomings are they exposing? What should you be mindful of when reading the book?

* * *

Action step

Do some research on the following books. For instance, look for the author's credentials, their political leaning, and their main arguments. Use Google, ChatGPT, or other similar tools as needed.

- *The 1619 Project, A New American Origin Story*, Nikole Hannah-Jones
- *The War on the West*, Douglas Murray
- *These Truths: A History of the United States*, Jill Lepore

D. YouTube content

YouTube is a wonderful source of information, but also of misinformation. Here are some questions to consider when watching content on the platform:

- **What is a fact, and what is an opinion?** Listen and practice sorting out the facts from the content creator's opinions. What claims are they making that may be only partially true or blatantly false?
- **Are they citing verifiable sources?** Do they show clear evidence and credible sources?
- **What's their expertise on the topic?** Did they study the topic extensively? Are they recognized by other people as experts in their field? In short, why should you listen to them?
- **Are they reasonable, or do they seek to polarize?** Are they being intentionally provocative? Do they oversimplify complex topics or polarize to generate views? Do they use the "us vs. them" strategy, targeting a "common enemy"?
- **What is emphasized, and what is omitted?** What topics are they choosing, and what topics are they omitting, and why? What people choose to talk about and ignore is, in itself, a bias and can say a lot about their motivations.

- **What are the main ideas of the content creators?** Look at the type of content they post. What main ideas are they pushing? How much of it is logical and factual vs. ideological (opinions, hearsay, anecdotes, personal experiences)?

As you get into the habit of asking these questions, you will become better at identifying misinformation.

* * *

Action step

Search for the following YouTube videos and criticize them by answering the questions below. Feel free to use ChatGPT, Google, or similar tools to help you with the fact-checking.

The Middle East Problem by Dennis Prager

You Will NEVER Eat Bread Again after Watching This by Dr. Eric Berg

Questions:

- What are facts, and what are their opinions or claims?
- Are they citing verifiable sources?
- What's their expertise on the topic?
- Are they reasonable, or do they exaggerate, oversimplify, or seek to polarize?
- What do they emphasize, and what do they omit?
- What are the main ideas of the content creator?

E. Conversations

We can learn a lot by having conversations with a variety of interesting people. However, conversations are often based on anecdotes, feelings, and personal experiences rather than on facts and data. Also, we tend to hang out with people who have similar beliefs to us, which makes it more likely for us to overlook facts and stay in an echo chamber. Ask yourself these questions:

- Who am I hanging out with the most?
- What do these people believe?
- What content are they following online?
- What topics are they talking about the most? Are their arguments based on science and logic?

* * *

Action step

- Make a list of the people you're spending the most time with.
- Write down what they believe in.
- Write down who they are following (health gurus, political figures, podcasters).

* Cholesterol Treatment Trialists' Collaboration. "Efficacy and Safety of More Intensive Lowering of LDL Cholesterol: A Meta-Analysis of Data from 170,000 Participants in 26 Randomised Trials." *The Lancet* 376, no. 9753 (2010).

* Knowler, William C., et al. "Reduction in the Incidence of Type 2 Diabetes with Lifestyle Intervention or Metformin." *New England Journal of Medicine* 346, no. 6 (2002): 393–403. doi:10.1056/NEJMoa012512.

* Wynder, Ernst L., and Evarts A. Graham. "Tobacco Smoking as a Possible Etiologic Factor in Bronchiogenic Carcinoma: A Study of 684 Proved Cases." *Journal of the American Medical Association* 143, no. 4 (1950): 329–336. doi:10.1001/jama.1950.02910390001001.

DIFFERENTIATING FACTS FROM INTERPRETATIONS

There are facts and there are interpretations. Sometimes we react to facts negatively, or we add our own biases to them, which leads to poor thinking and erroneous interpretations. To think critically, we must set our emotions and biases aside, look at facts, and consider the most logical interpretations of these facts. For the sake of practice, let's have a look at a couple of facts and see possible interpretations.

- **Fact #1.** There are more men than women in IT.
- **Fact #2.** There are more black people than white people in the NBA.

Now, here are some suggested interpretations.

Fact 1: There are more men than women in IT.

According to the World Bank, women account for only 28% of the students pursuing careers in information, communication, and technology (ICT). Now, why is that?

Below are possible interpretations:

- **Interpretation 1:** Women are actively being discriminated against.
- **Interpretation 2:** Men are generally more interested in IT than women.
- **Interpretation 3:** Society creates barriers to women who would like to study IT.

Our interpretation often dictates the actions we take. For instance, if we believe women are discriminated against or face societal barriers, we may put in place quotas. But if we think the reason is that men are more interested in computer science than women, we might decide against quotas. The truth might be that men are more interested in IT than women. After all, in the most egalitarian countries, women are still a minority in that field. Or it might be that women face discrimination or other challenges in those countries, too.

The point is that all three interpretations could be true. Or just one or two. Before drawing any conclusions, let's try to figure out the most likely explanation for such disparity.

Fact 2: There are more black people than white people in the NBA.

According to the 2023 NBA Racial and Gender Report Card, there are 70.4% black people vs. 17.5% white people in the NBA.

Below are some possible interpretations:

- **Interpretation 1:** White people are being actively discriminated against.
- **Interpretation 2:** Black people have natural talents.
- **Interpretation 3:** Society creates barriers to white men who would like to play in the NBA.

If we think white people are being discouraged from playing basketball, we may put in place specific measures to create an even playing field. But if we think black people have natural talents for that sport, we won't change anything. In truth, it's probably the result of multiple factors. For example, it could be cultural biases. Perhaps black people like basketball more. Perhaps they watch NBA games more often because they can identify with the players. Perhaps their genetics make them more suitable for that sport, and they naturally rise to the top as a result. Or perhaps basketball is more affordable or more readily accessible to black people.

I don't know the exact reasons. I'm just sharing hypotheses that would require further exploration.

To sum up, there is only one fact, but there are many potential interpretations. Before jumping to conclusions, we should take the time to consider all the possible factors and do more research.

* * *

Action step

Look at the facts below and write down possible explanations using your action guide. Note that the point isn't to find the right answer but to think for yourself.

- **Fact A:** In the US, women's life expectancy is 80.2, however for men it's 74.8 (according to the Centers for Disease Control and Prevention (CDC), as of 2022).
- **Fact B:** Car accidents occur mostly near home.
- **Fact C:** In Hong Kong, people consume a lot of meat, however they live to 85.8 on average.

AVOIDING KNEE-JERK THINKING

People sometimes have an automatic response upon hearing certain words. I call this “knee-jerk thinking.” This is usually the result of something they heard over and over in the media or from their friends. Let’s go over a few examples.

A. Electric cars catch fire and are dangerous.

When people hear the words “electric car”, they have an immediate reaction, often repeating what they’ve heard before. For instance, a common statement is that EVs catch fire more often than gasoline cars. In truth, EVs are far less likely to catch fire than gasoline cars^{*}. The misconception likely comes from the fact that when EVs do catch fire, they are more difficult to extinguish. In addition, it makes great headlines and generates more views (and more revenue) for news outlets.

B. Bitcoin is for criminals.

“Bitcoin” is another word that generates a lot of reactions. Upon hearing that word, someone’s immediate reaction might be, “Bitcoin is for criminals.” In reality, Bitcoin is a public blockchain where all the transactions ever made are recorded on the ledger. While a Bitcoin wallet isn’t associated with a specific person, there are sometimes ways investigators can track the funds. This is why the preferred option for criminals is cash, which is untraceable. According to a 2022 Chainalysis report^{*}, 0.24% of all crypto transactions were linked to illicit activities. Now, it is true that Bitcoin is often used for cybercrime, such as hacking or ransomware. However, for most crimes, such as drug trade, money laundering, or human trafficking, cash is by far the preferred choice.

C. You should buy, not rent.

If you rent, you’re paying for your landlord’s mortgage. You’ve likely heard that statement before. However, in some cases, renting might be a better financial decision. Here is why: when we buy, we must come up with the down payment and pay for the mortgage. In addition, we have to factor in the time and money we must invest in our place over the years. On the other hand, when we rent, there is no down payment, mortgage or any other costs associated with ownership.

By investing that money in an index fund and continuing to rent, we may end up wealthier. The point is, whether to buy or to rent depends on individual preferences as well as a variety of other factors. There is no simple answer.

What about you? When are you engaging in knee-jerk thinking? What do you believe that is simply not true?

To conclude this part, remember that facts don't care about your feelings. Avoid overreacting when you don't like the facts. Instead, strive to come up with interpretations that are as accurate as possible. Because if your interpretations are flawed, your resulting actions will also be flawed.

In the next section, let's see how to look at data more objectively so that we can identify facts and protect ourselves against misinformation.

* * *

Action step

Using your action guide, come up with at least one example of knee-jerk thinking you may engage in.

Then, investigate the following claims:

- The Great Wall of China is visible from space.
- Goldfish have a three-second memory.
- You only use ten percent of your brain power.

* AutoInsuranceEZ (2021), using NTSB/BTS data: ~25 fires per 100,000 EVs sold vs. ~1,530 per 100,000 ICE vehicles. Swedish Contingencies Agency (2023): ICE vehicles 20x more likely to ignite (0.08% vs. 0.004% for EVs).

* Chainalysis. "2023 Crypto Crime: Illicit Crypto Volumes Reach All-Time Highs." January 12, 2023. <https://www.chainalysis.com/ko/blog/2023-crypto-crime-report-introduction>.

PART III

LOOKING AT DATA OBJECTIVELY

These days we have access to an incredible amount of information. The biggest challenge is to separate good, reliable, useful data from bad, unreliable, and irrelevant data. Let's explore how to find more objective data that helps us achieve our goals.

AVOIDING INFORMATION OVERLOAD

Every minute, hundreds of hours of videos are uploaded on YouTube, and users consume over one billion hours of content daily. We're truly living in an age of information abundance. However, more information isn't always better. With so much content available, it's hard to figure out what's important (signal) from what isn't (noise).

A. The issues with information overload

With so much information readily available, there is always someone, somewhere, doing better than us. What we see online can distort our thinking, negatively affect our mood, and prevent us from setting realistic expectations for ourselves. More specifically, here are the main issues with information overload:

- **It makes us feel responsible for the state of the world.** The internet makes us aware of many of the tragedies happening around the world, whether they be natural disasters, wars, or crimes. While in the past we would have only been aware of what's happening in our village, nowadays our village is the entire world. As a result, we often feel responsible for issues happening on the other side of the planet. And because we can't do much about it, we feel powerless, anxious, guilty, and/or tormented.
- **It makes us feel insecure.** In the past, we were in "competition" with people in our village. With the internet, the "competition" has spread to the entire world. To make things worse, the internet tends to distort reality. Our social media feed is full of people who are richer, more attractive, or smarter than the majority of us. But they are also the ones who have received enough love from algorithms to appear on our feed. In short, they are anything but the norm—they are the exception. Yet, we can't avoid comparing ourselves with them. It creates unrealistic expectations and makes us feel inadequate.
- **It becomes harder for us to make decisions.** By being exposed to an enormous amount of information each day, we'll inevitably come across contradictions. That's not necessarily a bad thing, but it can make it

challenging to figure out what's true from what's false or what works from what doesn't. The content we consume daily is simply not tailored for us. It is fairly generic, which creates confusion. As a result, we end up being influenced by trends, jumping from one diet, investment opportunity, or relationship advice to the next, while failing to win the results we hoped for.

- **It leads us to lack perspective.** Constant exposure to a never-ending flow of information prevents us from zooming out and looking at the bigger picture. What's happening in front of us (i.e., what we see online) distracts us from the activities we should focus on to improve our life. Short-term thinking and immediate rewards replace long-term thinking and delayed gratification. In other words, everybody seems to be achieving results fast, so that's what we expect should happen to us. But we should remember that get-rich-quick schemes, cosmetic surgeries, and meme stocks are usually not the way to go to build a successful life. Amongst other things, success requires patience, strategy, commitment, and humility. These things neither come easily nor quickly.

B. How to separate signal from noise

A large majority of the information we consume is noise. That is, in the grand scheme of things, it's irrelevant. Put differently, noise distracts us from our goals. Signal helps us move toward our goals. Our level of success is often associated with our ability to reduce noise so that we can zero in on the information that matters (signal).

Below are some examples of noise:

- **Checking the stock market for hours each week.** If you're invested in the stock market for the long term, this isn't helpful.
- **Watching or reading daily news extensively.** Knowing what's happening around the world on a daily basis is not that useful. To make things worse, most of the content out there is negative.
- **Consuming a lot of information daily.** A large majority of information out there is nothing but noise. As we're looking to improve our diet, become fit, or build a business, we must select the information that helps us reach our goals and discard the rest.

C. How to find signals

Critical thinking is about finding signals and eliminating noise. The more tools we have to evaluate the quality of the information, the better we will be at extracting signals. Now, here is what you can do to separate signal from noise:

a. Clarify your goals

Building the life you desire requires knowing what you want. Without clear goals, you'll passively consume information instead of proactively seeking the information you need to attain your goals. By having specific goals, you prime your mind to look for the correct information and reduce the amount of noise you encounter each day. In short, goals enable you to increase signal and reduce noise.

b. Start with a high-level overview

Whether your goal is to learn a new skill, improve your thinking, or reach a financial target, avoid jumping on it too quickly. In other words, don't consume too much content. Otherwise, you will drown in a sea of information and will be unable to make any decisions or commit to a plan. Instead, seek to understand the big picture first. For example, ask AI software like ChatGPT to give you a brief summary of the topic you want to study or watch a couple of YouTube videos on the subject.

The point is, you don't want too many details early on or you risk feeling overwhelmed. Let's say you want to start running. Look for a credible content creator and watch a couple of videos for beginners. Then, start running. Later on, as you're running more consistently, you can look for more advanced content. This approach will prevent you from feeling overwhelmed.

One final tip: if you feel overwhelmed, it's a sign to stop. It's a sign that you have consumed more information than you can chew. Slow down. Zoom out. Refocus on what matters.

c. Invert your learning-to-action ratio

Learn less, do more. The more you do, the more feedback you receive from the real world, and the more you can adjust your actions and gain clarity. In other words, taking action helps you:

- Figure out what works and what doesn't,
- Uncover what you enjoy doing and what you hate doing, and
- Shed light on your natural talents and centers of interest.

Therefore, strive to take more action. It will help you gain clarity, which, in turn, will reduce noise and increase signal.

d. Consume less information

Content creators are on a never-ending hamster wheel, having to produce more content to grab people's attention, stay relevant, and make ends meet. To do so, they often play on people's fears, rely on clickbait titles, or seek to polarize. In other words, many content creators want to grab your attention. Don't give it away mindlessly. Instead, focus on what's useful to you.

e. Prioritize long-form content

These days, apps offer book summaries, telling you that you can acquire the knowledge from a book in less than fifteen minutes. The premise is that more knowledge is better. This is yet another manifestation of the consumer society we're living in. In truth, to develop critical thinking and acquire wisdom, we need much more than book summaries. We must immerse ourselves in the inner world of the author. We need to understand how a podcast host or guest is thinking. Put differently, we need to prioritize long-form content over short-term forms. Instagram reels or TikTok videos can be great sources of entertainment, but they are usually poor sources of learning.

Long-form content enables us to have a conversation with the content creator (in our head). We can follow their logic but also question their motives or challenge their assumptions. Could you imagine Einstein, Newton, or da Vinci spending their days on TikTok? Smart people read books and give themselves plenty of room to reflect. This is how they develop a more accurate model of reality and impact the world. If they do it right, that's how they learn to think for themselves and overcome many cognitive biases.

* * *

Action step

Complete the following exercise using your action guide:

1. For twenty-four hours, write down all the information you consume (YouTube videos, articles, books, podcasts, et cetera).
2. Next to each piece of each information write down whether it is signal (S) or noise (N).
3. Now, reflect on the following:
 - How much of the information you consumed is signal versus noise?
 - Are there times during the day when noise is more present?
 - How does noise affect your mood or ability to focus?
4. Finally, take concrete action to reduce noise and increase signal:
 - Allocate specific times during the day to check emails or social media.
 - Only subscribe to newsletters or follow accounts that provide useful information. Unsubscribe/unfollow from the rest.
 - Block time during your day to engage with content that aligns with your goals.

ESCAPING ECHO CHAMBERS AND CHALLENGING CONFIRMATION BIAS

While confirmation bias is a common bias, algorithms, by suggesting more of the same, amplify it and make us more close-minded.

A. The rise of algorithms

If you watch a video on a specific topic, YouTube will display more videos on the same topic. If you watch a few more videos on the same topic, your feed will become filled with such videos.

Let's say you come across a video promoting the carnivore diet. You'll then be recommended similar videos and might become convinced it's the optimal diet. In short, you become trapped in an echo chamber. However, the carnivore diet is controversial. Some people suffering from autoimmune conditions or chronic inflammation have reported seeing great relief. Others have been able to lose weight. For those people, the carnivore diet might be beneficial. But we don't have enough data yet to tell if it could be optimal for most people. On the other hand, the Mediterranean diet has been studied for decades, and we understand its positive effects.

The lesson here is, while algorithms can help us find information, they can also become a trap. YouTube, TikTok, and Instagram aren't designed to help us think, they are designed to hook us. In addition, to increase views, many content creators rely on sensational titles and seek to create controversy, often leaving nuances out or distorting the truth. The consequence is that many of us have become slaves to the algorithms. Instead of cultivating our ability to think and grow wiser as we age, we become more extreme with our diet, political views, or our religious faith.

Interestingly, algorithms exacerbate our tendency to belong, to be right, and to feel special. This is what makes them so irresistible. Once we find someone that speaks to us, we join their tribe (the desire to belong). Over time, it becomes part of our identity and we're reluctant to change our mind (the desire to be right). And belonging to a tribe makes us feel special—i.e., we know things that others don't (the desire to feel special).

Now that we understand how algorithms can erode our ability to think, let's see what we can do to protect ourselves.

B. Avoiding the trap of algorithms

To escape your echo chamber, here are a couple of things you can try:

- **Consume content from people advocating for and against something.** Proactively listen to people who have opposite views on a topic. It will help you build a nuanced viewpoint and prevent you from becoming extreme one way or the other.
- **Reconsider your views on topics that have no general consensus.** Take an inventory of your views on politics, economics, religion, fitness/diet, and any other topics that people regularly disagree on. Then, find people you disagree with. If you are a Republican, listen to Democrats. If you are an atheist, listen to believers. If you are on a carnivore diet, listen to people who promote a vegan diet (and vice versa). Keep an open mind and be willing to question your beliefs.

As you listen to people with a different opinion, you'll often find that you may have missed nuances. You may even feel confused, and uncertain of where the truth lies. However, understand that this is the price to pay when you become an independent thinker. Rather than seeking an illusory certainty, you must be willing to embrace the uncertainty of the world, no matter how unpleasant it may feel. That's how you know you're thinking for yourself.

C. The Influence of our social circle

The rise of algorithms isn't the only reason we get trapped in an echo chamber. Our social circle is another obvious one. As the saying goes, "We are the average of the five people we spend the most time with." This means that, among other things, we tend to hang out with people who hold similar beliefs to ourselves.

This helps explain why thinking critically is hard. Peer pressure and confirmation bias make it tough for us to challenge our beliefs. And if we question our beliefs, our new worldview and values may threaten the group cohesion, and our relationships may fall apart over time. This is something we usually don't want to experience. So, we'd rather keep believing what we believe—and keep our friends.

* * *

Action step

- Look at what you've watched on YouTube or other algorithm-dependent platforms.

- Now, for each topic, go watch a video that contradicts it. For example, search for videos featuring people with different political or religious beliefs. Or watch the videos of influencers that promote a radically different diet than yours.
- Observe how you feel. Are you easily convinced by their argument? If so, it means that you have the tendency to believe others. Or, conversely, do you think that what they say is nonsense? If so, perhaps you have become too entrenched in a particular set of beliefs.
- Think of the people you're hanging out with. Do they have similar beliefs as you? If so, how exactly? How are their beliefs affecting you?

PROTECTING OURSELVES AGAINST IDEOLOGIES

Intelligence isn't an antidote against stupidity. Otherwise, no smart people would ever believe in weird theories, join cults, or commit atrocious crimes.

Unfortunately, just because we read books doesn't mean we'll overcome our biases and become critical thinkers. In truth, we often let the material we consume influence us. Once we start adopting the worldview of the author, we look for more people with similar views. By seeking content that validates what we already believe, we create an echo chamber. Remember, Hitler, Stalin, and Mao were avid readers, but this didn't prevent them from being consumed by extreme ideologies that resulted in the deaths of millions of people. Instead of learning, these dictators sought to confirm what they already believed.

The danger of ideology

Ideology is what happens when we shield ourselves from the real world. It results from denying reality and ignoring facts that don't fit our narrative. Being an ideologue is the opposite of being a critical thinker. It makes us abandon logical reasoning for the comfort of believing in something that sounds good and makes us feel good.

Ultimately, ideologues are the enemies of truth. At some point in their journey, they discarded the pursuit of truth. So why do we become ideologues? Probably because we can't accept the fact that truth is unknowable. It makes us feel uncomfortable, restless, and powerless. Ideologies provide us with the reassurance we're looking for. They offer us clear answers and make the world appear simpler than it is. By choosing ideologies over truth, we don't have to think. We don't have to reevaluate our views on a regular basis either. Ideologies are empowering—or seen as such. They provide us with beliefs and a worldview to rely on. In an uncertain world, it makes ideologies attractive.

Ideology vs. reality

Ideologies cannot survive contact with reality. This is why ideologues constantly need to lie to themselves and to others. This is also why authoritarian regimes

must use propaganda. What is true doesn't need force. It simply stands on its own. On the other hand, what is untrue must be affirmed, over and over, until accepted as "true." For instance, the North Korean leader, Kim Jong Un, must rely on propaganda to convince his population that he is the supreme leader. Without propaganda and coercion, his population would see him as the ruthless dictator he is (and many probably see him that way as it is).

The key takeaway is, ideologies are not a good representation of reality. To think for ourselves, we must take a step back and try to understand what *we* think, not what others want us to think.

The point of learning

The point of educating ourselves isn't to accumulate more proof that our beliefs are correct. On the contrary, it is to look for evidence that may disprove our current thesis so that we can adjust our thinking. If we're unable to change our mind, we've abandoned critical thinking. Similarly, the real point of a debate isn't to win but to be willing to have someone poke holes in our thinking. Yet how often have you seen someone changing their mind or giving credit to the other person during a debate?

The bottom line is, we must keep thinking for ourselves. We must question our beliefs on a regular basis. This is how we can develop critical thinking and protect ourselves against the danger of ideologies.

* * *

Action step

- Choose a belief you hold strongly.
- Reflect on how you adopted this particular belief. What sources influenced you (e.g., parents, teachers, media)?
- Actively search for reliable information that challenges your beliefs. Read articles, watch videos, or listen to podcasts offering different views. Note any valid points that seem to contradict your belief.
- Observe how you react when encountering opposing views. Do you feel defensive, angry, or stressed out?
- Finally, assess whether your belief still holds.

AVOIDING BEING MISLED BY STATISTICS

Thinking for ourselves means understanding statistics properly so that we can avoid being manipulated. For example, as the economist Thomas Sowell explains, statistics regarding income can be misleading. When we think of the top one percent of earners, we tend to imagine a group of wealthy people at the top. However, the top one percent doesn't include the same people each year. That is, every year, many people move in and out of this group for various reasons:

- Entrepreneurs, athletes, or investors may have a few great years and reach the top one percent for a short time.
- People tend to make more money as they get older. They could end up in the top twenty percent or even top one percent of earners.
- People who sell their business may enter the top one percent for just one year.

According to a study from the US Internal Revenue Service, only thirty-nine percent of people in the top one percent of earners in 1996 remained in that group in 2005. Now, the point isn't to discuss inequality and how big of an issue it is, but to illustrate how statistics can be misleading. In that particular case, the error is to assume that the same statistics describe the same people year after year.

A. Statistics and percentages

Another way statistics can be misleading is by showing impressive results in percentage terms. A massive increase or decrease in percentage terms isn't necessarily meaningful. If I tell you my revenue increased by 500 percent this year, you might be impressed. But if I tell you it increased from \$1,000 to \$6,000, maybe not so much. Or let's say you manage to save fifty percent of your income but make \$1,000 per month. You're merely saving \$500. But if you make \$30,000 per month, you're now saving \$15,000 each month—or thirty times more.

Put it differently, percentages mean little if you don't know what the baseline is. It's easy to show incredible growth in percentages when starting from a low base,

but it's much harder when numbers are already big. For instance, it's harder for Apple to double its revenue than for a small startup.

Therefore, whenever you hear large percentages being thrown around, take a step back and look at the actual numbers. How many dollars are we talking about? How many more pairs of shoes, cars, or phones were actually sold? How many more people does it represent?

Statistics mean nothing without the proper context.

B. Statistics and trends

When looking at statistics, it's important to focus on the long-term trend, not just on a specific number or year. We need to have enough context to decide whether these statistics are useful or relevant.

For example:

- An investment fund may boast incredible returns over the past three years to lure new investors. However, luck alone can easily explain outsized returns over such a short period. Now, if they can show they have outperformed the stock market over the past twenty years, then it would be much harder to dismiss their results as mere luck.
- A country may struggle economically following a pandemic or financial meltdown. However, it doesn't mean that it is declining. We must look at the long-term trend (a decade or more) to get a more accurate picture of what's really happening.
- A new diet may help a lot of people lose weight within weeks. However, to assess the effectiveness of that diet, we need to observe whether the dieters can maintain that weight loss over the long term. It's possible that many participants will regain weight almost as fast as they lost it.

The bottom line is, a short-term trend gives room for luck, randomness, or temporary gains, while a long-term trend doesn't. Therefore, whenever you encounter statistics, zoom out and look at the bigger picture. People often distort statistics to have them say what they want to say. Don't be fooled.

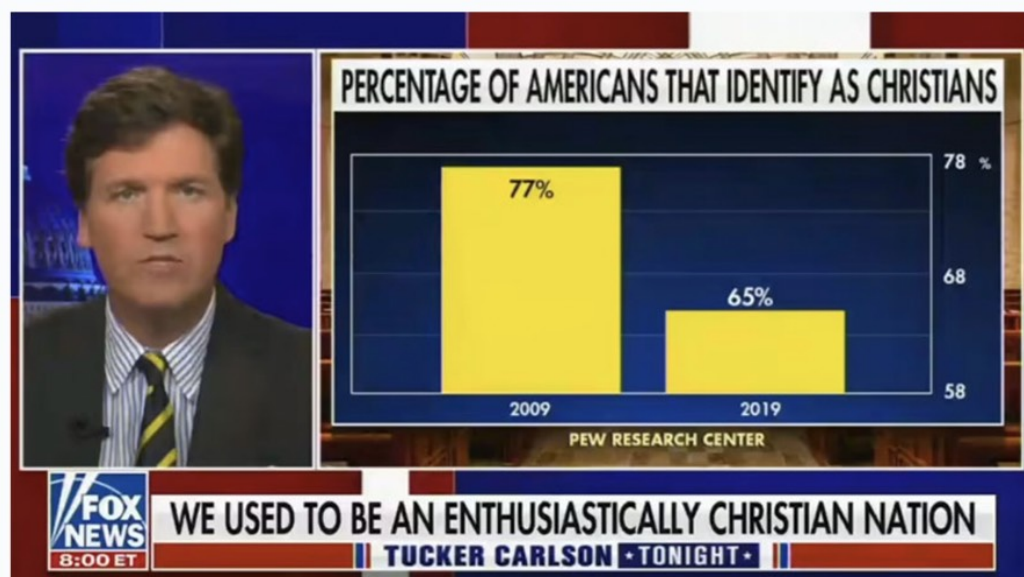
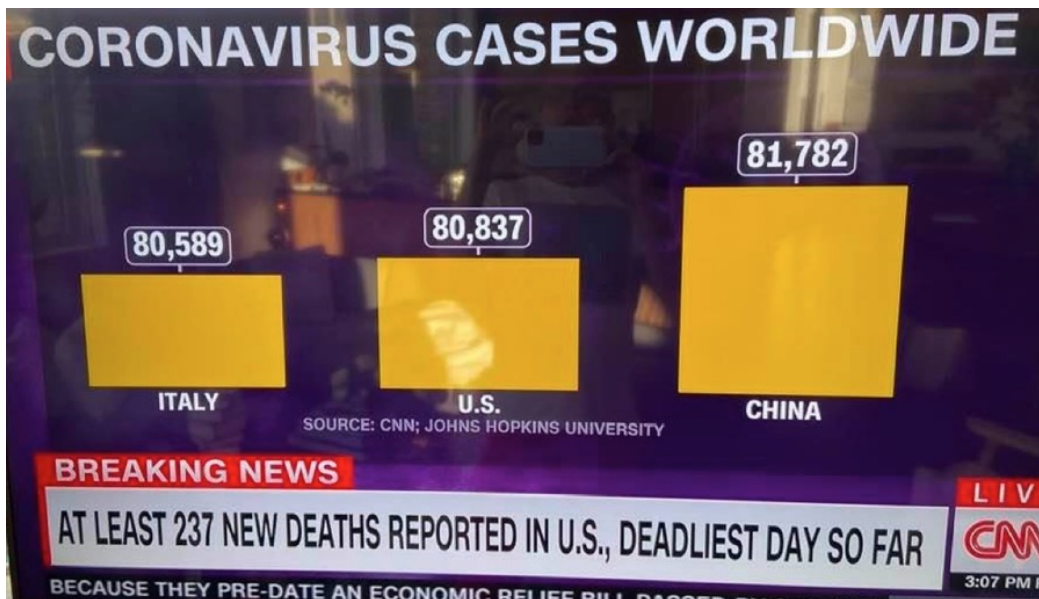
* * *

Action step

Colgate once advertised that "80 percent of dentists recommend Colgate."

While not a lie per se, this is misleading. Why could that be? Reflect for a moment and write your thoughts in your action guide (don't look for the answer online).

Look at the pictures below. Why are these graphs misleading?



Watch the YouTube video "[Misleading Graphs Real Life Examples](#)" by Prof. Essa that shows real-life examples of misleading graphs.

THINKING IN TERMS OF PROBABILITIES

A soccer team may have one to ten odds to win a game (over 90% chance) and lose. Meanwhile, someone may have a one in ten million chance of winning the lottery and win. The point is, just because an event is likely doesn't mean it will happen (and vice versa). While we can't reliably predict the future, we can assign probabilities to various events and take educated guesses as to outcomes. By understanding probabilities, we can make better decisions and improve our lives.

A. Avoid black-and-white thinking

Because a decision yielded good results doesn't mean it was the right one. Inversely, because a decision didn't bring the results you expected doesn't mean it was the wrong one. It's all about probabilities.

You may eat healthily and exercise regularly but still develop health issues. That doesn't mean your decision to eat healthily was wrong. Or you may choose a stable job over a risky position in a startup that ends up being successful. But it doesn't mean you made a mistake. In truth, you had no way of knowing the startup would be successful. Most startups fail.

Or, to give a personal example, I may spend months writing a book just to sell a few copies. It would be reasonable to think that perhaps the book wasn't good enough. But in reality, many factors explain the commercial success of a book. The book's quality isn't the only factor. Luck plays a big role, too. Readers happen to like a certain book. The cover works well for some reason. A celebrity shares it online and it goes viral. It could simply be that the timing is right.

The bottom line is, we shouldn't look at the outcome to assess whether we made the right decision. Instead, we should evaluate whether we made the best decision *based on what we knew at the time*. If so, we played the odds well. By making good decisions over and over, our life will most likely improve.

* * *

Action step

- Think of two or three decisions you made that didn't turn out the way you hoped for.
- Then forget about the results and assess whether the decisions themselves were good for you *at the time you made them*.

B. Case study on logical thinking and probabilities—"The Matrix Is After Me."

Whenever someone is criticized or attacked, they're quick to blame the system. They claim not to have done anything wrong. They claim to be the victim of a conspiracy. This may be true, but it's most often not the case.

Let's look at Andrew Tate, a former kickboxing champion turned internet personality, and use logical thinking and probabilities to evaluate some of his claims. Currently Andrew Tate faces various charges, such as human trafficking, rape, sexual intercourse with a minor, and money laundering, but has denied all the allegations. He argues that he is the victim of a political conspiracy. In his own words he said:

"The Matrix is trying to frame me, but God knows the truth. There's no evidence in my file, because I've done nothing wrong. Everyone knows I'm innocent."

"The Matrix doesn't want you to know the truth. They put me in jail because I'm teaching men how to be strong, how to be free. They can't control a generation of men who don't comply."

"The Matrix will only tell you what the Matrix needs you to believe."

In short, Tate claims to be a powerful individual who threatens the powers that be (i.e., the so-called Matrix). The establishment is after him because he challenges authority structures, promotes hyper masculinity, and advocates free speech—or so he says.

Now, let's pause for a moment and evaluate his claim. We want to figure out whether Andrew Tate is an actual threat to the "establishment". We also want to know whether the Matrix set him up or whether he's making things up. To do so, let's analyze the statements below:

- There is indeed a group of powerful people that controls the world (the Matrix).
- Andrew Tate has enough power to threaten them to the point where they must shut him down.
- He did nothing wrong, and all the allegations are totally unjustified.

For Andrew Tate's story to hold, each of these statements must be true. There must be a Matrix. He must threaten them. And he must be wrongly accused. Once we assess the likelihood for each statement to be true, we'll have a better idea of where to stand.

Statement #1. There is a group of powerful people that controls the world.

There are definitely powerful people in this world. The question is whether there is an elite group of people that colludes in order to control the world and push their agendas. This is possible but unlikely. The more likely scenario is that many groups of powerful people push their own agendas as opposed to there being one well-organized group pulling all the strings.

Statement #2. Andrew Tate has enough power to threaten the establishment.

Andrew Tate may have some influence due to his rise to notoriety and the few million dollars he might have accumulated. However, there is little evidence that indicates he has enough power or resources to impact world affairs or threaten the "powers that be."

Statement #3. He did nothing wrong, and all the allegations are totally unjustified.

Andrew Tate claims to have done nothing wrong, and he may well be innocent. But based on his statements and behavior, it's sensible to believe that the Romanian authorities have reasonable grounds for their allegations. If we think in terms of probabilities, which of the following options seems more likely?

1. The Romanian authorities charged him because they had some evidence.
2. They made up the case out of thin air because they saw him as a threat to the powers that be.

When looking at the situation objectively—and based solely on Andrew Tate's statements—I see someone who appears to manipulate women and men while selling dubious programs and promoting crypto scams. Let's review specific statements he made that seem to indicate this may be the case. These statements don't make him guilty, but they do enable us to question his narrative.

Indications that he manipulated women

Andrew Tate's statement below shows he was lying to women working for him to get a bigger chunk of the money they were making:

"Tax is also another important aspect of controlling your woman. You're not gonna pay anybody tax because you're getting paid in Bitcoin. But you're telling your girl that you pay tax. That allows you to do two things: one is another control element. Allows you to pay her a smaller percentage. So, for every ten thousand dollars they made, I would give them three and keep seven."

In addition, Andrew Tate explained how he would make women fall in love with him so that they would work for him. That's what he was teaching in his PHD program (Pimping Hoes Degree). In his own words, it was a program "that teaches basically how I got girls, how I met girls, how I got girls to like me, to fall in love with me to work on webcam for me because that's what I did. My ammo was, find girls, make them love me, and make them work for me. And that's how I got rich."

What Andrew Tate describes is similar to the "Loverboy Method"^{*}. Loverboys are human traffickers who usually operate by trying to make young girls or boys fall in love with them. Once they have the victims under their influence, they exploit them, for instance in the sex industry.

These statements seem to indicate that Tate was manipulating women.

Indications Andrew Tate manipulated men

In a [podcast interview](#)^{*}, Tate said:

"I had these guys selling their houses, life savings, loans, all of it to me. Give me it all."

Then, he added:

"We had this guy falling in love with my models, serious big time in love right, sending crazy money. And they were convinced they were going to meet the chick. This is where I kind of feel bad because they were like "can we meet? I've sent you \$200,000 can we meet? Can we meet? Can we meet?"

Furthermore, in a March 2022 article from the UK online news outlet, Mirror^{*}, Andrew Tate admitted that the whole business is a scam:

"It's a total scam. The model just has their hands on a keyboard that isn't even plugged in. I have real professionals who are fluent in English behind the scenes getting men hooked, finding out their interests, the name of their dog."

These statements seem to demonstrate he was manipulating men.

Indications he promoted crypto scams

Andrew Tate mentioned that he would never scam his audience by endorsing crypto^{*}. As he wrote on X (formerly Twitter):

"I DO NOT ENDORSE SHITCOINS. I'm not a scammer like every other 'influencer'. I don't need to rob my fans. I DO NOT HAVE ANYTHING TO DO WITH ANY CRYPTO."

In an interview, he also said that, "Crypto is the only scenario I can think of where you can make a whole bunch of money while benefiting society zero."

However, later on, he promoted multiple crypto currencies^{*}. On June 7, 2024, he wrote on X:

“Bought 10K of Fuck Madonna coin because Fuck Madonna never selling diamond hands what’s next @Madonna.”

Then, on October 19, 2024, he wrote:

“Having the power to randomly pump any coin on the chart is super fun.”

This shows a complete about turn and is another data point that indicates we should take what he says with a grain of salt.

To conclude, when thinking logically, we have no choice but to question any of the claims made by Andrew Tate. It doesn’t make him guilty. It just means that, based on his own statements, we can’t help but be skeptical.

Disclaimer: this analysis of Andrew Tate’s public statements and actions is based on verifiable evidence, including his own quotes and credible media reports. It aims to illustrate critical thinking principles and does not assert guilt in ongoing legal proceedings. All claims are presented as commentary on matters of public interest, not definitive judgments of fact.

In this section, we’ve explored traps that prevent us from looking at data objectively. We’ve discussed information overload and how it makes it hard to differentiate signal from noise. We’ve seen how, due to the rise of algorithms, coupled with our tendency to surround ourselves with like-minded people, we can end up in an echo chamber. We’ve explored the danger of ideologies and how being book-smart isn’t necessarily a good thing. We’ve uncovered how statistics can be misleading and why it’s important to think in terms of probabilities. Finally, we’ve dissected manipulative techniques and explained how to use logic and probabilities to guard ourselves against them.

In the next part, IV. **Protecting ourselves against manipulation**, we will explore in greater depth the way various people and organizations manipulate us and what we can do to protect ourselves and preserve our ability to think critically.

^{*} In their 2015 Situation Report on Trafficking in Human Beings in the EU, Europol noted: “Deception is commonly used to lure potential victims, including the ‘Loverboy’ method.” This indicates that the Loverboy method is recognized as a prevalent tactic in human trafficking, particularly for sexual exploitation.

^{*} These quotes come from an Andrew Tate’s interview reposted on YouTube: <https://www.youtube.com/watch?v=WKibLhAGP1A&t=243s>. The original source and publication date could not be verified.

^{*} Andrew Tate, quoted in “Brothers Make Millions Using Webcam Girls to Con Men Out of Money,” The Mirror, March 2022, <https://www.mirror.co.uk/news/uk-news/brothers-make-millions-using-webcam-26508739>.

^{*} Crypto News, “Finance News,” accessed June 1, 2025, <https://cryptonews.net/news/finance/28610563/>.

^{*} A non-exhaustive list of coins Andrew Tate promoted: \$VENOM, \$ROOST, \$FMADONNA, \$RNT, \$STOPG, \$G, \$STATE, \$DADDY, and \$TRW. Most cryptocurrencies he endorsed crashed over 90% post-promotion.

PART IV

PROTECTING OURSELVES AGAINST MANIPULATION

Not everybody has good intentions. Many people will try to exploit your desire to belong, to be right, to feel special, and your need for congruence. Let's see how they attempt to do so and explore ways to protect ourselves.

PROTECTING OURSELVES AGAINST FALSE EXPERTS

The world is uncertain, and we can't predict the future. That uncertainty makes us feel uneasy. This is why we tend to be attracted to and gravitate toward people who seem confident. Many influencers, self-proclaimed gurus, and so-called experts display a high level of confidence. They usually have strong opinions and a clear idea about the topics they talk about. One way they make themselves appear confident is by using words such as "always" and "never" and by generally speaking in absolutes.

The problem is that false experts lack nuance, which, paradoxically, is what leads many people to be drawn to them. Their sense of confidence and certainty is reassuring. The clear answers they bring make it feel as though they have everything figured out. But it often comes at a price: a distortion of truth and a black-and-white vision on many issues. Some examples of this are:

- **"Us vs. them" narrative.** Influencers claim that we are the good guys and that "they", whether "they" are Big Pharma, the global elite, atheists, or any other target, are the bad guys.
- **Good vs. bad.** Influencers argue that some things are always good and others are always bad. For instance, they might say you should never consume seed oils, that rent is always a waste of money, or that quitting your 9-5 job is the only way to be free.
- **True vs. false.** Influencers claim to know the truth, and they accuse people who disagree with them of being wrong. For instance, their political beliefs are the right ones, their religion is the correct one, or their diet is the best one.

This is an oversimplification of reality.

On the other hand, legitimate experts tend to be more nuanced. They acknowledge the complexity of the world and think in terms of probabilities, which is more aligned with how reality works. This is why they will often say words such as "it depends." They understand that some things may work for some people, but not

for others. They consider things on a case-by-case basis. As a result, rather than using “always,” they say “likely,” “often,” “possibly,” or “might.” And rather than using “never,” they use “unlikely,” “rarely,” or “only in some cases.”

My rule of thumb is to be wary of overly confident people. Here is one example: the geopolitical analyst, Peter Zeihan, when talking with Joe Rogan, stated confidently that Bitcoin would fall to zero^{*}. He said, “What’s Bitcoin at? \$16,000? It has another \$17,000 to go down.” The famous investor, Warren Buffett, also believed Bitcoin had no intrinsic value. He described it as, “probably rat poison squared.”^{*} And Buffett’s associate, the late Charlie Munger, labeled it as “worthless” and a “turd.”^{*}

Whether we think Bitcoin is the best technology ever or a giant Ponzi scheme is irrelevant. It is simply poor thinking to say it will go to zero without introducing nuances or presenting solid arguments. Charlie Munger was respected worldwide for his critical thinking skills. Ironically, he abandoned critical thinking when discussing Bitcoin. Instead of doing his homework, he dismissed Bitcoin as useless, probably because it broke his models or led to speculation.

The bottom line is, thinking for ourselves is hard. We must stay vigilant and remain curious. No one is immune to incorrect thinking and intellectual laziness, not even the smartest people on earth. Therefore, beware of people who seem overly confident, who have strong opinions on a variety of topics, and/or who seem to know it all. There is only so much one person can know. And bear in mind that if someone says something silly in one area, they are likely to say stupid things in others.

A. How false experts manipulate the truth

Truth is complex, so it is tempting for us to simplify it to make it fit our narrative. Charlatans tend to do the same thing. They use many techniques to distort the truth and get people to buy into a certain set of beliefs. Let’s look at common techniques they use:

a. Cherry-picking of data

By cherry-picking data, it is possible to build a case for almost anything. For example, there may be a consensus on a certain topic, with most studies pointing in the same direction. But we can pick the one study that aligns with what we believe.

A recent study showed a correlation between higher cholesterol levels and increased lifespan. It was quickly cited by some YouTubers as evidence that having a high LDL level isn’t a bad thing and could actually be healthier. However, in truth, there is a scientific consensus that high LDL is a risk factor for cardiovascular disease:

- According to the European Atherosclerosis Society (EAS) and the European Society of Cardiology (ESC), “There is no threshold below which LDL is

not atherogenic. Lower is better.”

- For the American Heart Association (AHA), “Reducing LDL-C is the primary target of therapy to prevent cardiovascular events.”
- The World Health Organization (WHO) recognizes high LDL cholesterol as a major risk factor for cardiovascular disease.

So, why did that single study show a correlation between higher cholesterol levels and increased lifespan?

Layne Norton, who holds a PhD in nutritional sciences, pointed out that such studies can be misleading as they fail to account for confounding factors. For example, older people often have low LDL levels because they are sick. Therefore, it's not surprising to see that they might not live as long as older people who are healthier, but also have higher cholesterol.

This shows how data can be misleading. We must analyze what lies behind the numbers rather than jumping straight to conclusions. Correlation isn't always causation. What seems logical at first sometimes isn't.

b. Anecdotes

Anecdotes are another example of manipulating data. As additional data points, anecdotes can have some value, but when taken in isolation, they hold little weight.

Here are some examples of anecdotal evidence:

“I know a guy who cured cancer by fasting, therefore, rather than doing chemotherapy and other cancer-related treatments, that's what you should do.”

“I'm making \$20,000/month with a multi-level marketing company. Therefore, if you join the company and do exactly what I do, you'll become wealthy, too. Success leaves clues. You just have to follow my path.”

“Grace Jones and Agnes Fenton lived to 112. They both drank whiskey daily. Therefore, if you drink a little whiskey each day, you might live to 100.”

The problem with anecdotes is that they sound appealing while having little to no value. When we see other people achieving the results we want by following a specific method, we think, “If it works for them, it will work for me.”

In many cases, it's just not true.

Manipulators rely on plenty of anecdotes to convince us to join their cult, believe in their fringe theories, or buy their products. And that's when they aren't downright making things up. Learn to spot anecdotes. As a rule of thumb, any claim that isn't backed up by studies (and usually several studies) is merely an anecdote. It can point in the right direction, but it shouldn't be taken in isolation.

c. Illusion of expertise

Another common issue is when the wrong experts give their take on a topic. For instance, just because someone has “MD” after their name doesn’t mean we should believe everything they say in relation to medicine. Medicine is a vast field, and one doctor can’t know everything. This is why there are many specialties (neurosurgery, pathology, ophthalmology, et cetera). For example, an ophthalmologist is not the most competent person to talk about cholesterol. The same logic applies to other fields. A mathematician isn’t a climatologist, and a fitness influencer isn’t a nutritionist. People may not always be ill-intentioned but take what they say with a grain of salt when they venture beyond their field of expertise.

d. Doubt mongering

Even if 99 percent of the scientific community agrees with a conclusion, just one lone wolf who disagrees can spread doubt. This is especially true when considering people’s dislike of Big Pharma or lack of trust in the establishment. As a result, a consensus can quickly be put into question. For instance, in the case of cholesterol, we just need one cardiologist with enough visibility online (by appearing on a major podcast, for example) to make people doubt. Manipulators have figured out that they can gain influence by spreading doubt and being nonconformist.

e. False analogies

Gurus and self-proclaimed experts like to make erroneous comparisons to manipulate us. They often compare apples to oranges while trying to have you believe that their analogies make sense. False analogies can bypass our critical thinking and mislead us if we aren’t careful. Below are a few examples:

“Everything is a pyramid.”

When MLMs are accused of being pyramid schemes, they sometimes reply by saying everything is a pyramid. They argue companies have a pyramid-like organizational chart with a CEO, executives, managers, employees, et cetera. Or that your family is also a pyramid with great-grandparents, grandparents, parents, and children.

This is obviously a false analogy used to deflect criticism. Just because something has a pyramidal structure doesn’t mean it’s a pyramid scheme. What makes something a pyramid scheme is when people make money from bringing in new members rather than from selling valuable products or services.

“It can’t be wrong because it’s not illegal.”

Just because something is legal doesn’t mean it’s ethical. Slavery was legal for centuries. Child labor was legal during the Industrial Revolution. And advertising cigarettes to children used to be legal as well. These days, we could argue that many advertising techniques or schemes are unethical while being legal. It’s up to us to decide if what we do is ethical based on the consequences our actions may have.

“If we’re being lied to about one thing, then they must be lying about everything.”

Spreading doubt is a powerful way to get people to buy into various dubious theories. But just because a government messed up one time and tried to cover up their mistakes doesn’t mean they lie all the time. Yes, it’s a red flag, but we should evaluate people’s claims before jumping to conclusions.

“Just like Galileo was persecuted for being right, so are modern whistleblowers.”

Putting oneself in the position of the martyr is a common tactic used by manipulators. This specific example assumes that being rejected by the mainstream is proof of being correct, which is logically false. In truth, many ideas are rejected by experts for a reason—they have no merit and aren’t supported by science or data. When someone is ostracized by the scientific community or by the media, it’s likely because they promote ideas or ideologies that aren’t backed by science and that may even be harmful.

f. Oversimplification

No one can grasp the complexity of the world. But at the same time, we want certainty. This is why we find simple solutions to complex problems appealing. It gives us the illusion that we have found the missing piece of the puzzle. The more vulnerable we are, the stronger our desire for certainty and personal agency tends to be.

Manipulators offer “solutions” to our biggest problems. Conspiracy theorists may point the finger at a group of powerful people. Manipulators claim that certain people may be controlling the economy, enslaving us through work, making us sick through 5G antennas, and so on. They are the reason for our problems. Unscrupulous marketers may tell us that there is a magic formula to becoming rich (e.g., by day trading, joining their MLM companies, or buying their courses).

In short, manipulators claim that problems always have a simple cause to which they bring an obvious “solution”. YouTube’s thumbnails often mention one trick, secret, or mindset that will change everything for us. It makes us believe that there is something we’re missing. I call it the “magic pill syndrome.” In truth, many factors will determine the achievement of our goals, such as strategic thinking, trial and error, self-awareness, continuous experimentation, humility, persistence, patience, a constant desire to learn, and of course, some luck along the way. There is no magic pill.

The bottom line is, beware of people who claim to make things simpler than they are. They probably have their agenda and are trying to sell you something, whether a product, a service, or an ideology.

g. Appeal to authority

Manipulators appeal to authority to make their claims more credible. They may use quotes from famous people, often taken out of context (and sometimes fake), to make it seem as though they are saying something important. They may also rely

on (fake) testimonials to boost their credibility. They can't be the ones telling you their product is great. You won't believe it. But if they can get dozens of people to say it for them, now you might believe it.

Notice whenever someone is trying to appeal to authority to enhance their influence, they usually have something to hide—or to gain.

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UNDERSTANDING THE MECHANISMS USED BY CONSPIRACY THEORISTS AND GURUS

We can all get sucked into conspiracy theories, be tricked by scammers, or fall under the influence of gurus. Our critical thinking can always be attacked, and our ability to think for ourselves should not be taken for granted. In this section, we shall explore various mechanisms used by cult gurus and other manipulators.

A. Why we join cults, believe in conspiracy theories, or fall for scams

We often fall prey to manipulation when we're offered something that we really want. Manipulators understand what we desire most and make us believe we can have it. We just have to do as they say.

As the saying goes, everybody wants something. So, what do people want? They want to belong, be right, or feel special. They also want to feel as though they understand how the world works and what causes all their problems. And they may want more money, freedom, happiness, or power. When they can't get those things anywhere, they sometimes become victims of cults, fall for get-rich-quick schemes, or believe in weird conspiracies.

To sum up, we join cults or believe in conspiracies because they fill a void within ourselves. Now, let's see in more detail what techniques gurus use to manipulate us.

a. Emotional vulnerability

There are several elements that make people prone to manipulation. A major one is emotional vulnerability. People who buy into get-rich-quick schemes, get sucked into weird theories, or join cults often go through difficult times. They may have lost a loved one, suffer from depression, or hate their job. Feeling unable to escape their situation, they look for a way out.

Now, you may think that you are too skeptical to join a cult or be scammed. But you could find yourself in a vulnerable position. In addition, cults exist in various degrees. Some cults are obvious, but many cult-like organizations act more sneakily. The point is, we aren't always as smart as we think. The first step to protect our critical thinking skills is to acknowledge that fact.

b. Cults and the BITE model

Steven Hassan, an expert on cults and mind control, has created the BITE model. According to this model, cults rely on various manipulative techniques that fall into four categories:

- Behavior Control (B).
- Information Control (I).
- Thought Control (T).
- Emotional Control (E).

Let's look at each of these so that we can better understand how cult or cult-like organizations operate.

Behavior control (B)

Cults regulate their members' behavior. They might decide what followers should wear, control their relationships, or punish them when their actions are deemed unacceptable.

For example, in the 1970s, the cult leader, Jim Jones, instructed his followers when they should work, eat, and sleep. In addition, they were required to listen to long sermons and engage in communal labor, which left them with little autonomy or time to think.

Information control (I)

Cults also try to control the information their followers receive. They may discourage them from consulting outside sources in addition to promoting misinformation or using propaganda. By controlling the information, they can impose their narrative and prevent critical thinking. For instance, the Church of Scientology labels critics as "Suppressive Persons" (SPs) and forbids members from reading any material that criticizes the organization.

Thought control (T)

Cults also attempt to control their members' thoughts. They may instill a specific belief system or teach thought-stopping techniques—i.e., psychological strategies that prevent critical thinking and quiet doubt (e.g., mantras, meditation, or prayers).

For example, the Unification Church tells members to pray if they begin questioning the leadership. Members of the International Society for Krishna Consciousness chant "Hare Krishna" to eliminate intrusive thoughts. And members of the cult group Heaven's Gate repeatedly said, "I'm overcoming human ways," to remove doubt.

Emotional control (E)

People are less likely to leave if they are emotionally attached to a group. This is why cults use various techniques to keep followers under their influence. A common one is “love bombing.” New members are repeatedly told how great they are, which makes them feel good about themselves. As a result, they are less likely to leave and more likely to comply later on. On the other hand, they may also be criticized or punished when they question the way things are done. This approach is similar to an abusive relationship, in which one partner manipulates the other by telling them how much they love them while, at the same time, continuously berating, shaming, and insulting them.

To sum up, cults try to control their followers’ actions, sources of information, thoughts, and emotions. As a result, cult followers can lose their ability to think critically and lose themselves in the process. This is why, after they leave a cult, it can take them years to process what has happened to them.

B. The Matrix narrative and conspiracies

I love the movie *The Matrix*, but the metaphor has been overused, often to manipulate people. The narrative is that there is a malevolent force controlling society and enslaving us and that we must escape it to finally be free. It’s a compelling story that ill-intentioned people use to manipulate their audience. For instance, that force may be:

- **Your 9-to-5 job.** Regular jobs are scams. You are a slave to your company. Your freedom has been taken away, and you’re barely scraping by. You must escape the Matrix by becoming yet another “entrepreneur.” Are we to believe that there should be no employees, then? No waiters, cooks, store clerks, nurses, police officers, et cetera?
- **Big Pharma.** Pharmaceutical companies want to keep you sick. That’s how they make profits. You must take the red pill and discover how to reclaim your health. It opens the door to many charlatans who try to sell you snake oil and other miracle products. Are we then to believe that every single employee working for Big Pharma is part of a big conspiracy?
- **The reptilians.** Shape-shifting reptilian aliens are controlling the world. You must elevate your frequency and reclaim your power to break free from their control. Are we to believe that alien reptilians exist and are manipulating us while operating in the shadows?
- **The New World Order.** There is a cabal that pushes its own agenda and controls the world. You must fight against them to reclaim your freedom. How likely is it that a coordinated group of people has enough power to decide the fate of the world and influence the entire population?
- **Circles of Satanist pedophiles.** There is a global elite that engages in satanic rituals, child trafficking, and pedophilia. They are the reason why there is so much evil in this world. While Satanists and pedophiles do exist, how likely is it that there are highly organized rings influencing the world affairs?

The bottom line is, the Matrix narrative is often used by gurus and conspiracy theorists who seek to push an agenda and manipulate us. It allows them to cast doubt on our vision of the world and instill fear. By introducing doubt, they create room in people's minds to establish a new vision of the world. In other words, if they can make people believe that their model of reality is wrong, they can start reprogramming their mind. It all starts by creating doubt and instilling fear.

EVALUATING CONSPIRACY THEORIES

A conspiracy theory is a claim that a secret plot by powerful groups is behind a certain event or situation—often without verifiable evidence to support it. In this section, we'll see how to approach conspiracy theories and assess whether they might be valid. Once you understand how to evaluate them, you can use the same framework for other conspiracy theories. For the sake of example, we'll analyze the following conspiracy theories:

1. Moon landing hoax.
2. Shape-shifting reptilians.
3. Flat Earth.

A. Moon landing hoax

Some people believe that we never went to the moon. After all, if we had been there, why haven't we gone back since?

Now, how should we approach conspiracy theories such as the moon landing hoax? The best way is to look at probabilities. We can point out many "coincidences," but if each of these coincidences has a near-zero chance of probability, they are likely to have no merit. In short, instead of letting coincidences affect our judgment, we would be better off looking at the probability of various events to be true. To do so, we can ask ourselves the following question:

"What would need to be true for that conspiracy to be true?"

By identifying the necessary conditions for a conspiracy to be true and assigning probabilities, we can better assess its odds. For instance, below are a few statements that would need to be true for the moon landing hoax to be real:

- There were credible whistleblowers.
- The Soviet Union lied about their ability to monitor Apollo missions in real-time.

- The moon rocks are fake, and the independent labs that confirm their authenticity are part of the conspiracy.

Now, let's break down the probabilities of each of these events.

a. There were credible whistleblowers

When analyzing a conspiracy theory, we can look at the number of people involved. Generally speaking, the more people involved, the less likely it is for the hoax theory to be true. This is because human beings are terrible at keeping secrets. With enough people involved, sooner or later, whistleblowers will come forward.

Dr. David Robert Grimes from Oxford University developed a mathematical model that calculates the probability of a conspiracy being exposed over time based on the number of people involved^{*}. The Apollo mission involved around 400,000 engineers, scientists, and technicians. According to Grimes' model, for the moon landings to be a hoax, it would have been exposed within 3.68 years. Now, what is the likelihood that these whistleblowers would have stayed silent for over fifty years?

According to that model, it's 0 or close to it.

In fact, many real conspiracies have leaked with fewer people involved. For example, the 1972 Watergate scandal involved only a few dozen people, but it was exposed in less than two years. In 2013, Edward Snowden, a former NSA contractor, exposed multiple government surveillance programs run by the NSA, including PRISM. That program started in 2007 and involved thousands of people. Yet, it was exposed only six years later.

The bottom line is, the more people involved in a secret project or conspiracy, the likelier it is to leak sooner or later.

b. The Soviet Union lied about their ability to monitor Apollo missions in real-time.

The Soviet Union had huge incentives to prove that the moon landing was fake. If it was a hoax, they would likely have figured it out. Yet they tracked Apollo missions in real-time with radar and radio telescopes. So, what is the likelihood that the Soviet Union lied about the tracking?

It's near zero.

For the sake of this demonstration, if the Soviet Union did lie and the moon landing never happened, the following things would need to be true:

- The US faked the moon landing.
- The USSR agreed to go along with that lie.
- All Soviet space scientists, engineers, and intelligence officials remained silent for decades.
- No KGB defectors spoke out.

- Modern Russian space agencies keep covering it up.

The probability for each of these events is also near zero. But when you combine them together, it becomes extremely low. Let's assume that the probability for each is one percent and that they're independent. Then the likelihood that the Soviet Union lied would be as follows:

$1\% \times 1\% \times 1\% \times 1\% \times 1\% = 0.00000001\%$ or 1 in 10 billion

In short, it's highly unlikely.

The issue with coincidences

Conspiracy theorists stack up coincidences to build their case. Sometimes they are right; there is a conspiracy. But more often than not they fall prey to biases. For instance, we have seen that it is unlikely for the Soviet Union to have lied. This observation doesn't go away just because we show coincidences or gather evidence. That is, we still need to account for the 1 in 10 billion chance that the Soviet Union lied about monitoring the moon landings. Believing that the pictures taken on the moon were fake or that we didn't have the technology to go to the moon doesn't magically erase these odds.

c. The moon rocks are fake, and the independent labs worldwide that confirm their authenticity are part of the conspiracy

NASA distributed moon rocks to 135 countries, and independent labs worldwide analyzed them, confirming they were real. What is the likelihood that the moon rocks are fake and that all the labs are part of the conspiracy?

Moon rocks were independently analyzed by labs around the world*. Their geological composition matches what we expect from lunar material. For the conspiracy to hold, all the scientists working in these labs would have to be complicit. This is highly unlikely.

Overall, based on the probabilities we gave for each of the events above, it appears statistically unlikely that the moon landing was a hoax.

Now, let's delve into another conspiracy theory.

B. Shape-shifting reptilians

According to the conspiracy theorist, David Icke, shape-shifting reptilian aliens secretly control the world. These reptilians have ruled the world for thousands of years. Some famous reptilians include Queen Elizabeth II, George W. Bush, Bill and Hillary Clinton, and the Rothschild family.

Now, how likely is that conspiracy theory to be true? Let's identify all the events that must be true for that theory to be confirmed. We can then assess the odds. These events are as follows:

- Aliens must exist.
- They need to be reptilians.
- They control the world without being detected by anyone.
- They are able to shape-shift.

Without going in detail, we can say that the odds are extremely low. Conspiracy theorists may craft a compelling narrative by relying on coincidences, anecdotes, or ancient myths, but it doesn't change these odds.

Let's look at one more example.

C. Flat Earth

In the 2021 POLES survey, ten percent of participants agreed with the statement that "the Earth is flat," while an additional nine percent were unsure^{*}. In a 2024 survey conducted in Brazil, around twenty percent of the participants agreed with that same statement^{*}. More broadly, it is estimated that between three and ten percent of the world's population believes that the Earth is flat. Now, for the flat Earth theory, the following conditions are necessary:

- Every space mission and image of Earth has been faked.
- Every satellite image and private space mission photo is fake.
- Gravity as we know it does not exist, which implies all physics experiments are incorrect.
- Pilots, airline companies, and GPS providers are covering it up.
- It's possible to reach the edge of the Earth.
- No astronaut, scientist, or government official has leaked proof.
- Every country on Earth has agreed to hide the truth.

What is the likelihood that all these conditions are met?

Again, it's almost zero.

While the moon landing hoax, the shape-shifting reptilians, or the flat earth theories may be far-fetched, we can apply the same logic to any conspiracy theory. By identifying what would need to be true and assessing the odds, we can dismiss many conspiracy theories with a fairly high level of confidence.

Conspiracy theories and patterns of beliefs

David Icke doesn't just believe in reptilians, but also in a wide range of other conspiracies. For instance, he claims that the moon is a hollowed-out spacecraft used by aliens as a command center to control Earth. Now, not only do reptilians need to exist, but the moon needs to be hollowed out. This demonstrates a specific pattern of belief, which weakens his claims and makes his theory even less plausible.

The point is, conspiracy theorists tend to believe in not just one, but multiple conspiracies. It shows that these individuals, rather than being driven by a desire for truth, cling to specific patterns of belief. These patterns are likely the result of several factors, such as:

- A lack of trust in the government.
- Insufficient critical thinking.
- An excess amount of stress.
- Paranoid tendencies.

Let's remember that two improbable conspiracy theories are unlikely to both be true. And if we find ourselves believing in several conspiracies, it likely says more about us and our biases than about the level of truth of our beliefs. So, let's stay vigilant and question our beliefs. Some conspiracies may turn out to be true, but most won't.

Incompetence or nefarious Intent?

We tend to believe that people have nefarious intent when, in reality, they may simply be incompetent. When we think a large group of people is conspiring behind the scenes, it's important to take a step back and ask ourselves whether:

- there is really such a large group of organized people.
- they really have malicious intent.

In truth, instead of a well-organized group executing its evil plan, it might be a bunch of incompetent people acting in their self-interest. In short, the issue is often incompetence, not malice. Conspiracy theories give too much credit to the ability of large groups of people to cooperate in secret to execute their nefarious plans.

In addition, we should consider Occam's razor's principle. According to this principle, when several explanations are possible, the correct one is usually the one that requires the fewest assumptions to be true. In other words, if an explanation is too complicated and requires many unlikely events to occur, it's unlikely to be true. Here are some examples of Occam's razor:

- If you have a headache, it's more likely to be due to tension or dehydration than to be a symptom of a rare neurological disorder.
- If you can't find your keys, it's more likely that you forgot where you put them than someone stole them.
- It's more likely that the pyramids were built using forgotten techniques than they were erected by aliens.

In short, the simple explanation tends to be the correct one because it's much more likely. Let's start there before considering more complex explanations.

When conspiracy theories are true

We've spent time warning against the danger of conspiracy theories while showing many of them are likely false. To be fair, we also have to acknowledge that a few may actually be true. Being a critical thinker means being skeptical of conspiracy theories, but it also means being willing to look at facts and data and draw our own conclusions. Below are a couple of conspiracy theories that turned out to be true (at least in part).

MK-Ultra mind control program

Project MK-Ultra was a CIA program that aimed at developing mind control techniques. Conspiracy theorists claimed that the CIA was experimenting with LSD and mind control on civilians, which was widely mocked. Later on, it was revealed that the CIA did administer hallucinogens to unknowing subjects, including prisoners, hospital patients, and regular citizens*. Conspiracy theorists also claimed that the government was torturing people using drugs and psychological techniques. This has also been confirmed*. Many people were indeed subjected to extreme techniques, such as sensory deprivation or electroshock therapy.

Not every claim made by conspiracy theorists were true though. For instance, their claim that the MK-Ultra program was tied to satanic rituals has not been supported by any released documentation.

Watergate scandal

In 1972, several individuals working on President Nixon's reelection campaign were arrested during a break-in at the Democratic National Committee headquarters. Later on, investigations revealed deliberate efforts to cover up Nixon's involvement. Facing impeachment, Nixon was forced to resign in 1974.

Before the scandal was exposed, some conspiracy theorists and investigative journalists raised serious concerns but were, for the most part, dismissed or ignored. For example, they claimed that:

- The White House orchestrated the break-in to spy on political opponents.
- Senior officials tried to conceal the operation.
- Nixon's administration misused agencies, such as the FBI and the CIA, to delay investigations.
- There were secret recordings of Oval Office conversations.

All of these allegations turned out to be true. So, as you can see, not all conspiracy theorists are wrong. On some occasions, they may be right, at least partially.

The bottom line is, be skeptical of most conspiracy theories while understanding that there may sometimes be elements of truth. Whenever necessary, do your own research by asking yourself what would need to be true for a conspiracy to hold.

Key points

Whenever you hear about a conspiracy, ask yourself how likely it is to be true. To do so, consider the following:

- **What would need to be true.** If, for the conspiracy to be true, it requires a number of unlikely events, then it's unlikely. Remember that the probability of two independent events with a ten percent chance of both happening is merely one percent ($10\% \cap 10\%$).
- **The number of people who would have to be involved.** If, for a conspiracy to be true, it must involve thousands of people, it's unlikely to be true. The more people are in the know, the likelier it is that whistleblowers would have come forward.
- **The fact that pieces of "evidence" or coincidences don't cancel out the odds.** For instance, the absence of stars in photos taken on the moon doesn't change the fact that the Soviet Union did monitor Apollo missions.

If you get sucked into a conspiracy theory, zoom out and ask yourself what would need to be true for that conspiracy to be accurate.

Alongside conspiracy theories, another trap that millions of people fall for is the promise of wealth and freedom made by certain companies. That's what we're going to explore in the next section.

* * *

Action step

Try to confirm or invalidate one of the conspiracy theories below using what you have learned in this book:

- QAnon
- Chemtrails
- 9/11 Inside Job
- Vaccines and Autism
- 5G and mind Control

Remember to ask yourself the following questions:

- What would need to be true for this conspiracy to be true? Make a list using your action guide.
 - How likely is each of these events to be true? Give it a rough probability.
 - What are the main pieces of evidence that conspiracy theorists rely on? How strong/weak are they?
-

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EXPLORING MLMs

Multi-level marketing (MLM) is a business model where people sell products or services directly to consumers and earn commissions not only from their own sales but also from the sales made by people they recruit and by the members those people recruit (hence multi-level). Members are encouraged to recruit people, as their income depends on both their sales and the sales of their recruits.

MLMs are often sold as a way to become financially independent and break free from the typical 9-to-5 job. Prospects are told that such businesses offer unlimited income potential. The sky is the limit. The harder they work, the more money they can make. However, only a few people at the top ever succeed.

A. The problem with MLMs

The issue with MLMs is that the business model is flawed. It's designed so that a majority of people lose money. And the few who make money usually make less than the minimum wage when factoring in the number of hours they work each week. Jon M. Taylor, PhD, in his report, "The Case (for and) against Multi-Level Marketing," has demonstrated that about 99 percent of people who join MLMs end up losing money*. When 99 percent of people across an entire "industry" lose money, it is clear that the business model doesn't work. In the end, it's a near zero-sum game in which people at the top accumulate almost all the money while making the people at the bottom believe that they can become rich (or at least make a side income by working part-time).

a. Why MLMs don't work

If we're generous, we can say that MLMs rely on a flawed business model. If we're honest, we must arrive at the logical conclusion that they are scams. Why? Because people who run them know what they're doing. They understand that the majority of participants will lose money while the few people at the top will enrich themselves. Yet they choose to build that "business". If you think I might be exaggerating, consider how you would answer the following question:

If you had the opportunity to create a business knowing that it would enrich yourself and a few others while leading 99 percent of people who join to lose money, would you do it? And would you consider it ethical?

b. The fundamental issue with MLMs

The fundamental problem with MLMs is the fact that members take a cut of the sales generated by anyone below them (i.e., members they recruit, members those members recruit, and so on). Now, why are they entitled to part of their work? They didn't make the sales nor create the product. They may not even have interacted with many of the people below them. But they get their cut. MLM people call it passive income, I call it time theft. This is one of the biggest problems with this so-called business. Time is our scarcest resource. Stealing other people's time is one of the worst things we can do. But that's how MLMs work.

c. Why people join MLMs

If MLMs are such a losing proposition, how come millions of people join them each year? Why do they fall for them? Why do they waste time trying to "be their own boss"? There are a couple of reasons for this.

First, most people simply don't understand what MLMs are and how they work. Some people are uneducated. Others are simply young and a little naive or greedy. Yet others are in vulnerable situations. When they are told that they can make a lot of money and are sold an unrealistic dream, they are willing to give it a shot. But if they understood the math, they would stay away.

Second, they are being lied to. MLMs rely on deception and manipulation to attract new members. Telling the truth—that a majority of people lose money—just doesn't make for a great pitch. Consequently, their members have no choice but to lie in order to recruit more people and make money. You may have seen large-scale conferences where thousands of people are clapping, jumping around, and hyping each other up. On stage, speakers tell their success story while showing their lifestyle and bragging about how much money they make. But this is merely an example of survivor bias. The few who make it are put on display to give the illusion that everyone else can. This is one of many ways MLMs lie. They make people believe that if they work hard and follow the blueprint, they can become rich, too.

As a result, people buy into an impossible dream and waste their time and energy. Instead of making money and reaching financial freedom by seizing a "business opportunity", they lose money, become trapped in a manipulative system, and miss real opportunities to advance their career and improve their life.

B. Are MLMs cults?

What is not based on the truth requires constant lies, manipulation, and convincing to continue existing. MLMs are based on lies—the lies you can make passive income, be your own boss, or make a lot of money. That's why, to survive,

they must manipulate people and distort the truth. This is also why they tend to be cultish. Members must be convinced to stay.

To simplify the way the MLM journey unfolds, we can look at three different stages.

The initial stage—the promise

To recruit members, MLM people, like cult leaders, must make a compelling offer. There must be an enticing promise, a dream worth pursuing, an opportunity that can't be missed. In the same way engineers may aspire to join SpaceX to build cool rockets, you want people to get excited about joining your company. The problem is that, while regular companies may offer a genuine opportunity to do meaningful work or develop new skills, or they may provide you with a good paycheck or nice perks, MLMs have little to offer. This is why they must sell a dream. And for you to buy this dream, it must be compelling enough and touch upon the things you desire most such as:

- Being able to work from anywhere you want.
- Having the freedom to set your own schedule.
- Making a lot of money.
- Going on fancy vacations and buying cool things.
- Building passive income.
- Not having to work too hard.

This is why MLMs are manipulative. They're built on false promises that anyone a little naive, greedy, or vulnerable may very much want to believe (even when something feels off).

The second stage—personal development

Because MLMs are structurally flawed, members need to be manipulated so that they won't leave. For instance, MLM people may love-bomb new members by continuously praising them and saying how amazing they are. At the same time, if members fail to achieve results, they will often be told that it's their fault. They simply didn't work hard enough or weren't committed enough.

That's where MLMs and personal development meet. MLMs understand that under the guise of personal development, they can further manipulate members and increase their level of commitment. To gaslight their members they emphasize the importance of cultivating the following qualities:

- **Desire.** You're not successful because you don't want it badly enough. You need to strengthen your desire. (They may recommend you read *Think and Grow Rich* by Napoleon Hill, for instance.)
- **Personal Responsibility.** You are responsible for everything in your life. If you're not getting the results you want, it's because you're blaming

others, and you aren't taking full responsibility (not because the "business model" is flawed).

- **Grit.** You should never give up. If other people are successful, you can be successful too. It's just a matter of time.
- **Positivity.** You need to stop letting negative people and naysayers steal your dreams. If your family members and friends aren't rooting for you, you should distance yourself from them. In some cases, MLMs may tell you to cut ties with your best friend or even break up with your partner.

Desire, personal responsibility, grit, and positivity can be wonderful assets when pursuing the right goals. However, when taken in the context of MLMs, they become tools to gaslight people and pretend everything is their fault when they fail. Let's remember that 99 percent or more of people who ever join MLMs lose money. Yes, you could "win," but how likely is it? And at what price? How many people will have to lose for you to win? How many lies will you have to tell? What manipulative techniques will you have to use?

To help members build the right "mindset," MLMs regularly invite motivational speakers and other self-help gurus to speak at their rallies. Speakers get compensated generously, while MLMs can motivate their troops. This is a win-win relationship. Over the years, many major personal development figures have spoken at MLM conferences*. For instance, Robert Kiyosaki, author of *Rich Dad Poor Dad*, which has sold over thirty million copies, has been a speaker at various MLM events.

The third stage—more lies and manipulation

In the final stage, you either reach a point where you want to leave, or you become so entrenched with the MLM that you're evangelizing everybody you can, using the same techniques you fell victim to.

If you're lucky, someday you realize you have to leave. That's when your upline (the person who recruited you) starts using all kinds of manipulation techniques on you. They tell you to try harder, invite you to one more seminar, or even suggest that you leave your partner because they aren't supportive enough*. But it doesn't work this time. For some reason, you now see through the lies.

This may not exactly be what people in MLMs experience, but it's not uncommon. Some people may join for a short period and leave without any significant negative impact on their lives. Others will stay for years and lose their friends, sometimes even family members, and, more importantly, themselves. When they finally manage to leave, it will take them months to process what has happened to them (not unlike cults).

C. MLMs and cult-like manipulations

MLMs may not necessarily match the exact definition of a cult, but some MLMs use similar manipulation techniques. For example, many of them seek to condition the minds of their members with "transformational," "motivational," or "mind

training” courses. The point of most of the manipulation techniques is to prevent people from thinking critically. Let’s look at some examples.

a. Clichés

Clichés may seem harmless, but they are the enemy of thinking. While truth is nuanced, clichés remove nuance and offer a simplistic vision of the world. Their frequent use erodes our critical thinking. Here are examples of widespread clichés in MLMs:

- “Winners never quit, and quitters never win.” → This prevents you from questioning whether you’re going in the right direction. If you are part of an MLM, you should actually quit because the model is flawed.
- “Success is ninety percent mindset.” → This makes you think that something is wrong with you, that you don’t believe hard enough to succeed. Mindset is important, but only when your goal is attainable, not when you’re sold a lie.
- “If you’re not making money, you’re not working hard enough.” → This makes you think that you’re to blame because you didn’t work hard enough instead of realizing the business model doesn’t work.
- “Be your own boss.” → This makes you believe that you own your business and that you are responsible for the results you get (or fail to get). In truth, in an MLM, you aren’t your own boss. You are, in fact, a contractor.
- “Job = Just Over Broke.” → This discredits 9-to-5 jobs while making you believe that you will make far more money by joining an MLM, which is statistically false.
- “You have to spend money to make money.” → This justifies constant purchases, like starter kits or monthly quotas to keep your rank (and enrich people at the top).

b. Love bombing

Love bombing is when new members receive excessive praise and attention. They feel special as a result, which makes them more vulnerable to further manipulation. For instance, new members may be told some of the following:

- You’re already doing better than most people do in their first week. You’re amazing.
- The moment I saw you, I knew you were a natural leader. We need more women like you.
- I don’t say this to everyone, but I can tell you have that energy we look for.

c. Affirmations and mantras

In MLMs, people are often encouraged to listen to motivational tapes or repeat affirmations as part of their daily routine. While these may be useful when trying to reach our goals, when used in the context of MLMs, they become a huge waste of time that leads people to keep going for months or years because, “winners never quit, and quitters never win.” In a sense, affirmations and mantras can be seen as a way to internalize clichés, which, over time, undermines critical thinking.

To sum up, MLMs, like cults, use many techniques to get people to stop thinking for themselves. Let’s stay away and strive to warn people around us. Our time is precious, and we shouldn’t waste it by pursuing empty promises.

To learn more about MLMs, watch Robert L. FitzPatrick’s [The Cult of Multi-Level Marketing](#) or read his book, [Ponzinomics](#)^{*}.

* * *

Action step

Watch the following one-minute video from the MLM company Herbalife at the following URL: <https://www.herbalife.com/en-us/business-opportunity>.

- Write down key phrases, promises, and emotional appeals used by the presenter. Are they making claims about unlimited income, financial freedom, et cetera?
- Consider how you feel when watching the video. Do you find it appealing? Do you understand why others would be interested in joining?
- Based on what we discussed in this section, can you see what the problems are with this type of business?

^{*} Taylor, Jon M. ‘The Case (for and) against Multi-Level Marketing: The Complete Guide to Understanding the Flaws – and Proving and Countering the Effects – of Endless Chain “Opportunity” Recruitment’. Consumer Awareness Institute, 2011. Available at: <https://www.scribd.com/document/149677221/THE-CASE-FOR-AND-AGAINST-MULTI-LEVEL-MARKETING-by-Jon-Taylor>.

^{*} Personal development figures who spoke at MLM include Jim Rohn, Tony Robbins, Zig Ziglar, Les Brown, Brendon Burchard, John C. Maxwell, or Brian Tracy.

^{*} Surviving Quixtar – Amway’s Secret Alter-Ego | Multi-Level Misery. In the video, Kevin, who was a member of the MLM company Amway, explained how his upline suggests he leaves his fiancée. At 20min40s, “I would never forget this for as long as I live. Because this f*** guy, who I don’t know from Adam, tells me, my fiancée that I’ve been with at this point for three years, we just got engaged, he says to me, “if she can’t support you then you probably shouldn’t be with her.””

^{*} FitzPatrick, Robert L. ‘Ponzinomics: The Untold Story of Multi-Level Marketing.’ Fitzgerald Publishing, 2020.

VARIOUS SCAMS AND WHY WE FALL FOR THEM

According to Merriam-Webster, Cambridge, and Oxford dictionaries, a scam is “a fraudulent or deceptive act or operation, typically designed to trick someone into giving up money or valuable information.”

A scam almost always suggests a compelling promise. The scammers need to offer something people want in order to attract them.

A. Why do we fall for scams

We often fall for scams because of weaknesses that scammers are quick to exploit.

- **We want to make money (greed).** Having money is connected to many different values, such as freedom, status, or power. When we have money, we have an almost limitless number of options. We can buy the things we want, take a break from work, travel, or use money as a way to boost our self-esteem. But making money is hard, which is why we want to believe that there are easy ways to earn it.
- **We want a shortcut (laziness).** Any meaningful goal usually requires an extraordinary amount of time and effort, whether it is losing a lot of weight, building a thriving business, or becoming an expert in any field. Because it's hard and there is no guarantee we'll ever succeed, we're constantly looking for shortcuts.
- **We trust others too much (naivety).** We sometimes trust people too much. We want to believe people are right because it would make our life so much easier. So, we fall for scammers even when we know their offer is probably too good to be true.
- **We're too complacent or don't know enough (ignorance).** Sometimes, we don't know enough about a topic, so scammers can charge outrageous prices or make us believe that something will be easier or faster than it really is. This leads us to have unrealistic expectations that scammers can then exploit.

B. Examples of scams

Below are some examples of scams.

a. Law of attraction, abundance, and manifestation

Law of attraction gurus know how to attract money to themselves—usually, your money. One of my acquaintances, who was into the law of attraction, scammed me to the tune of thousands of euros. He needed money to run ads for his course and told me I would double my investment if I lent him money. I was naive and a little too greedy, so I did. I never saw that money again. A few months later, he came knocking at my door to beg for more money. He needed to pay his rent or would be evicted. He was so insistent that I ended up “lending” him more money. Eventually, he was evicted, and he never paid me back.

Of course, this is just one anecdote. But many people in the law of attraction community are faking it. They hope that one day, they will make it. After all, that’s what they preach: visualize what you want, believe it hard enough, and you’ll receive it—but this is usually not the way life works.

The point is, stay away from anything related to the law of attraction, abundance, or manifestation. Visualization can have its place, but it’s only one of many components required to reach our goals. The problem with the law of attraction is that it relies on unfalsifiable claims. That is, if you apply it and it doesn’t work, gurus will blame you. They’ll say you didn’t believe hard enough or your feelings weren’t positive enough to attract the things you want. In short, it’s always your fault but never theirs. Note that there is no scientific evidence supporting the law of attraction.

In the end, making money is about having useful marketable skills or offering valuable products or services. Unfortunately, the law of attraction won’t help much with that.

More broadly speaking, many religious or spiritual gurus manipulate people by making unfalsifiable claims and blaming their audience when they don’t get the promised results. If their followers achieve positive results, the gurus take credit for those, if they fail, it’s their fault. In the end, the gurus always win.

b. Online entrepreneurship

Making money online sounds great on paper. And many people promise you they can help with that. For whatever reason, an online business appears to be some kind of magical way to print money. It’s as if you could manifest money out of thin air. However, the truth is, an online business is still a business, and online customers are real people who must be convinced to spend their money on your product or service.

Yet gurus would like us to believe that we can make “passive income,” that making \$10,000/month online is easy, or that we don’t need any skills to build a successful online business. We just have to follow their recipe, whether it’s trading forex, creating an automated YouTube channel, or launching a course that will bring thousands of dollars each month for the rest of our life. If making

\$10,000/month was easy, nobody would work in factories, retail stores, or any other difficult jobs that require long hours while paying little.

Building a successful online business, like a traditional business, takes a lot of time and effort. People pretending otherwise are lying to you.

c. Investment opportunities

Some people want to take your money so that they can enrich themselves. They may promise you a high interest rate if you invest your money in their fund or cryptocurrency. But, often, when the promise seems too good to be true, it's because it usually is.

Years ago, I put cryptocurrencies on a platform called Celsius. It was a big platform with around half a million users and looked legit. The Caisse de dépôt et placement du Québec (CDPQ), one of the largest pension fund managers, even invested \$150 million in October of 2021. But in July 2022, Celsius filed for bankruptcy, and I lost the majority of the money I had on their platform*. The interest rate they offered seemed reasonable (around four percent per year for Bitcoin). But they had to get that money somewhere, which entailed trading cryptocurrencies or lending them with interest. When their plan didn't work as expected, they engaged in fraudulent activities, such as using new deposits to pay earlier investors.

The bottom line is, you can't get rich quick. And an investment that yields interest always involves some risk.

C. How to avoid falling for scams

Unfortunately, most people will get scammed at least once in their lifetime. In this section, we'll go over a few things you can do to protect yourself against scammers.

Ask yourself whether it's too good to be true. As the saying goes, if it seems too good to be true, it probably is. If someone promises you fast results, easy money, unrealistic returns, or cheaper prices than anywhere else, stop to think. If it's not an outright scam, there are probably hidden risks. There is no free money and there are no positive results without effort.

In doubt, do your own research. If something seems just a little too good to be true or if you have any doubt at all, do some research about the product or service you're about to buy, the person you're thinking of hiring, et cetera. Search online for authentic reviews, ask your friends their opinions, or compare with similar products.

Never buy anything on the spot. Always wait at least 24 hours before making any important purchase or life decision. If you are pushed into making an instant decision, back off. It's almost never the case that a real opportunity will present itself only once or be available for just 24 hours. Most of the time, it's just overly

aggressive marketing. Don't spend your money or give your money to anyone without stopping to look at the situation in detail and with a critical eye.

Be aware of emotional manipulation. Some of the hardest scams to avoid are the ones that hijack our brain and create enough fear to prevent us from thinking. For instance, recently, my girlfriend's dad received a phone call from a woman pretending to be his daughter. She told him that she was in an accident and had broken ribs and her mouth was bleeding. She was at the hospital and needed money immediately. Other examples may be emails saying you haven't paid your taxes, that your account has been hacked, or that your bank information has been stolen. When you feel threatened or scared, give yourself a little time to calm down and think. Emergencies are rare. If someone is pushing you to act fast, stop for a moment and assess the situation objectively.

Don't give out your personal information. Scams may be an attempt to steal your money directly but also indirectly. This is the case with phishing, for instance. Whenever someone asks for your password, credit card number, or bank details, be immediately suspicious. Often, people will try to pose as legitimate entities, such as banks, insurance companies, or government offices. They may send emails with fake logos, use convincing language, or pretend that your account will be suspended if you don't verify it now by sending the information requested. As a rule of thumb, avoid clicking on links in emails. Instead, go directly to the official website and double-check to make sure you typed the right URL.

Understand that scams can be extremely sophisticated. Sometimes, the reason we get scammed is that we don't realize how sophisticated scams can be. A good way to protect ourselves is to know examples of elaborate scams for future reference. Scams follow specific patterns; when we know these patterns, our alarm bells should ring, which can give us just enough time to stop and think. This is especially important when we are emotionally manipulated (scary email, phone call from someone stating that a family member has been in an accident, et cetera). Here are a few examples of elaborate scams:

- **Business Email Compromise (BEC).** Hackers take over a business email account, usually from a CEO, CFO, or another high-ranking executive. Then they send emails to employees in the finance department and request wire transfers. For instance, in 2016, an aerospace parts manufacturer lost 42 million euros when an employee followed the instructions sent by his "CEO."
- **Impersonation of CEOs Using AI Voice Cloning.** Scammers use AI to deepfake the voice of a CEO. Then they ask employees to transfer funds. In 2019, using this technique, scammers managed to get an executive to wire them \$243,000.
- **Tech Support Scams.** Scammers purchase Google Ads for terms like "Microsoft Support." When people click on the ads, they're redirected to a fake site with a phone number. When victims call the number, they're convinced to install remote software on their computers.

Remember, we can all get scammed. So, let's be prepared.

Case study—sightseeing in Bangkok

It's your first day in Bangkok, and you decide to explore the city. While walking toward the Grand Palace, a local informs you that the site is closed due to a special holiday. Bad luck! But they suggest other temples to visit and offer a tuk-tuk ride at an affordable rate.

The driver takes you to a few temples. As you walk toward one of them, a stranger on a bench strikes up a conversation. He tells you that today you can purchase gems at a discounted price. He just bought some. You thank him and continue your journey.

After visiting a couple more temples, you're taken to a jewelry store. The store looks nice, and you're reminded that you can indeed buy jewelry stones today at a discount. Salespeople encourage you to buy gems, explaining that you could resell them for a profit back in your country.

Do you buy a gem? Why or why not?

In this section, we've discussed how to protect ourselves against manipulation. We've seen how to spot false experts, analyzed how to approach conspiracy theories, and explained why we fall for various scams. In the final section, let's discuss how to identify common patterns and cycles so that we can think more logically and make better decisions.

* * *

Action step

Think of a time when you encountered a scam. If you can't recall one, search for a current scam online (e.g., law of attraction coaching, crypto pump-and-dumps, "make \$10K/month online" courses). For instance, research the Ponzi scheme, OneCoin.

Answer the following questions in your action guide:

- What was promised? (e.g., easy money, fast results, no effort required).
- What emotional trigger was used? (e.g., fear, greed, urgency, exclusivity).
- What tactic made it seem legitimate? (e.g., professional design, testimonials, use of authority figures).
- What part of you was it targeting? (e.g., greed, laziness, naivety, ignorance).

Then write down:

- What would a skeptic notice right away?
- What were the red flags?

Now, write down one or two rules you will follow from now on. Below are some examples:

- "If I feel like it seems too good to be true, I walk away."
- "I never spend money after the first pitch. I wait for at least 48 hours."
- "I research at least three independent sources before committing to anything."

* "Canadian Pension Fund loses \$150 million in Celsius investment bet", Finbold, July 21, 2022. <https://finbold.com/canadian-pension-fund-loses-150-million-in-celsius-investment-bet/>

PART V

IDENTIFYING PATTERNS

To think critically, we must also become experts at recognizing patterns. The more we understand patterns, the better equipped we will be to make sound decisions. In this section, we will go over key patterns that we should be aware of.

CONTINUITY BIAS

We tend to project the present into the future, believing the future will be similar. This is called the “continuity bias.” However, the future will be different—and probably radically so. This is especially true today as we enter an age of disruptive technologies driven by artificial intelligence.

Let’s look at a few concrete examples of how the continuity bias has led people to make poor predictions.

A. Population growth vs. food production

In the 1970s, some experts believed that the world was on the brink of massive famines and that hundreds of millions of people would die from hunger. The main argument, advanced by Malthus a couple of centuries before, was that food production grew linearly while the population grew exponentially. Therefore, the theory suggested that unless we limited population growth, many people would die from famine. In reality, thanks to many innovations, we are able to increase food production, avoiding large-scale famines. The lesson here is that we shouldn’t predict the future merely by projecting the present into the future. Instead, we should try to model future innovations, even though these can often be unpredictable.

B. The internet

Few people could have imagined the impact that the internet would have on our lives. The astronaut, Clifford Stoll, in an article published in *Newsweek* entitled, “The Internet? Bah!”*, doubted the internet would ever replace traditional newspapers. In 1996, *Time* magazine wrote, “Remote shopping, while entirely feasible, will flop—because women like to get out of the house, like to handle the merchandise, like to be able to change their minds.” In the end, the internet became far more disruptive than most people imagined. And there was no way to predict exactly how the technology would evolve and what people would use it for.

C. The airplane

In the early 1900s, many experts believed that it would take a very long time before a plane could fly. In 1900, Simon Newcomb, a mathematician and astronomer, doubted a plane could fly with the technologies of the time. On October 9, 1903, the *New York Times* published an editorial entitled, “Flying Machines Which Do Not Fly,”^{*} in which they joked that it might take between one and ten million years before a plane could fly successfully. Yet two months later, the Wright brothers made the first successful flight with a heavier-than-air machine.

The point is, we tend to underestimate the impact of new technologies and how they can dramatically and rapidly change our lives.

* * *

Action step

List three assumptions you have about the future. For instance:

- “I will keep the same job for the next ten years.”
- “Real estate will always go up in value.”
- “AI won’t affect my industry.”

For each assumption, imagine one disruptive scenario that could radically change that future and write them down. Here are some examples:

- A new technology automates your job.
- Real estate collapses because of population decrease.
- AI offers your service at 10 percent of the cost.

For each disruption, identify one positive opportunity that could emerge if such a change were to happen.

Finally, think of a time in your life when something you believed impossible or unlikely did happen. What lessons can you learn from that experience?

^{*} Stoll, Clifford. “The Internet? Bah!” *Newsweek*, February 27, 1995.

^{*} “Flying Machines Which Do Not Fly.” *New York Times*, October 9, 1903.

EXPONENTIAL GROWTH

Once upon a time, a king, who was impressed by the invention of chess, decided to reward its inventor. The inventor didn't want anything fancy. All he asked for was one grain of rice for the first square, two for the second square, four for the third square, doubling each time so as to fill all the 64 squares of the chessboard. The king obliged. But what appeared to be a small reward turned into something far bigger. The few grains of rice quickly grew to 524,288 on the 20th square, 537 million on the 30th square, and over a trillion grains on the 40th square. On the 64th square, the total number of grains reached 18 quintillion. This is far more than the entire world's rice production. To put this number into perspective, 18 quintillion is roughly comparable to the total number of grains of sand on all the beaches on Earth. This story illustrates the power of exponential growth and how counterintuitive it can be.

Now, why is it important to understand exponential growth? When we think linearly, we risk missing opportunities and failing to grasp the major changes that lie ahead. For instance, we could miss investment or career opportunities. Thinking for ourselves requires that we grasp how things may evolve in the future so that we can make the most of potential opportunities and trends.

Here are additional examples of exponential growth.

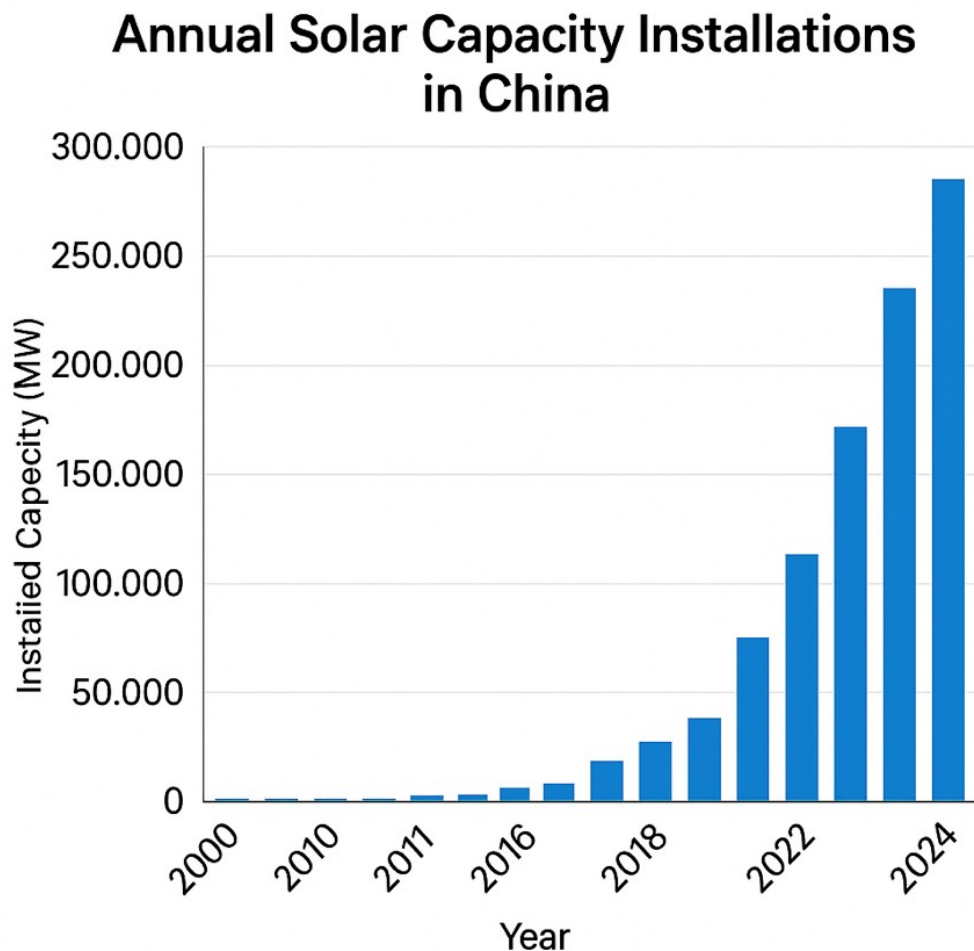
A. Compounding

Imagine you started investing \$150/month at age eighteen. If we assume a seven percent annual return, you would have made 7.5 times more profits than if you had started at age forty (\$577,203 versus \$77,220). That's the power of compounding and it demonstrates how wealth can grow exponentially after a certain point.

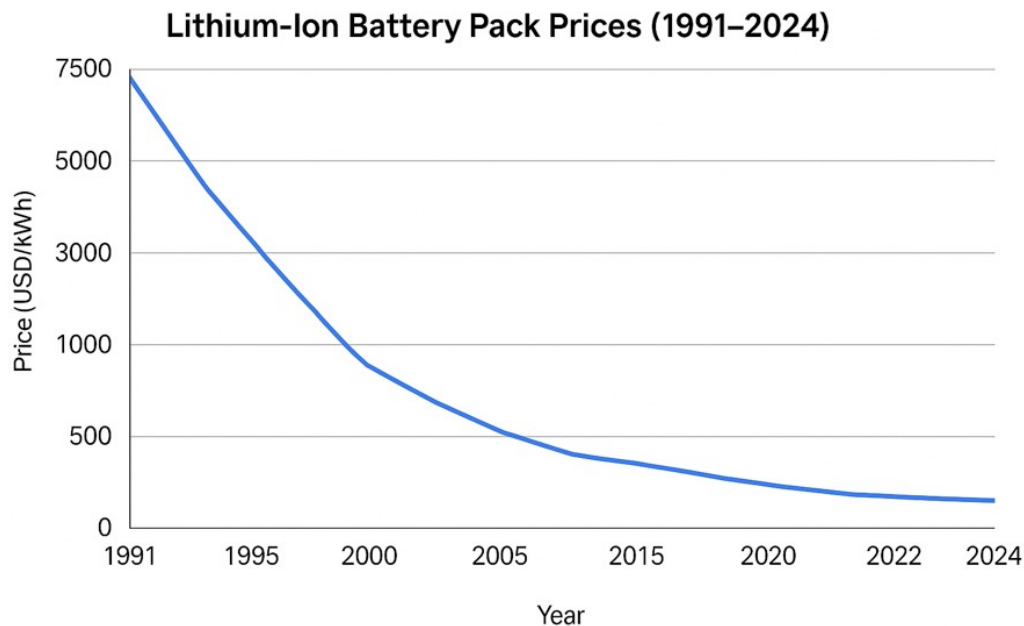
Starting Age	Invested (\$)	Final Amount (\$)	Profit (\$)
18	\$84,600	\$661,803	\$577,203
40	\$45,000	\$122,220	\$77,220

B. Cost decline

We often fail to understand that the cost of new technologies can decline drastically. One example is solar energy. For many years, solar panels were deemed too expensive to be a viable scalable option. However, the cost per watt of solar panels has decreased exponentially, from \$76.67 in 1977 to \$0.25 in 2023 (a 99.67 percent decrease)*. As a result, China has ramped up its solar power installations significantly in recent years, from generating only 16 MW in 1999 to 277 GW in 2024. (Note that this is only the price of producing the panels. The cost is much higher when taking into account installation.)



We can notice the same trend with lithium-ion battery packs. The cost dropped dramatically from \$7,500/kWh in 1991, when the technology was nascent, to \$115/kWh in 2024 (a 98.47 percent decline) and is expected to decline further as the technology matures*.



C. The scalability fallacy

“It won’t scale” is a common argument that results from people’s inability to think exponentially. It stems from people not recognizing the creativity, resourcefulness, and ingenuity of humans while failing to grasp how exponential growth works.

I remember the early years of the internet when it would take minutes just to load a low-resolution picture. At that time, most people didn’t believe we would be able to watch videos online. Now, hundreds of millions of people enjoy streaming services in high definition. Consider the first computer. The ENIAC had a speed of 0.1 MHz and 200 bytes of RAM and weighed thirty tons. In 1945, if you told people that in the future we would have a far more powerful computer in our pockets, they would have laughed. Today, an iPhone 15 Pro has a speed of 3.7 GHz, 8 GB of RAM, and weighs 6 ounces. Put differently, an iPhone is 37 million times faster and has 40 million times more memory while being 160,000 times lighter and around 7,000 times cheaper than the ENIAC. Similarly, when the first car was put on the road, people didn’t imagine that, within a relatively short period, horses would completely vanish from the roads.

These days, we hear similar arguments. Batteries won’t scale. The electric grid won’t scale. We’ll never have functioning humanoid robots in our lifetime. Autonomous cars will take decades to hit the roads. Let’s be careful before making

predictions. New technologies tend to be slow at first but accelerate exponentially, often surprising a lot of people along the way.

Thinking for ourselves requires that we integrate exponential growth into our mental models so that we avoid being swayed by wrong narratives and make poor decisions.

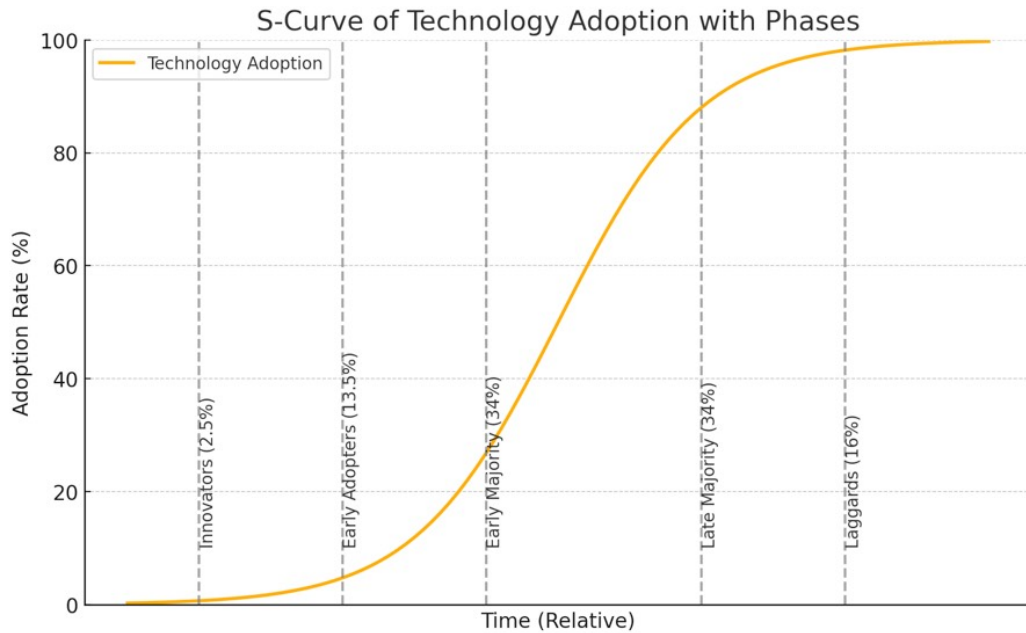
D. Wright's Law

Wright's Law states that whenever we double production, costs decrease by a constant percentage. This was theorized by Theodore P. Wright in 1936. While studying the production of airplanes, he noticed that the labor cost to produce an airplane declined by ten to fifteen percent every time production doubled*. Since then, a similar effect has been observed in many other products. The cost reduction results from factors such as economies of scale, automation, technological improvements, and so on.

For instance, this law has been observed in the production of lithium-ion batteries. The cost of producing such batteries has decreased by about twenty percent with every doubling of production. As a result, their cost has fallen by over 98 percent compared to the 1990s. The same logic applies to solar panel production, 3D printing, aerospace, and other technologies.

The key point is that while costs drop significantly at first, once a technology has reached maturity and billions of products have been made, cost decline slows down. Wright's Law explains why. For example, EVs have become cheaper in the past few years and will continue to get cheaper. EVs are fairly new, and the battery technology and manufacturing processes will continue to improve as production ramps up. It also explains why internal combustion engine (ICE) cars likely won't get cheaper. Car companies have been manufacturing ICE cars for over a century and have produced billions of them. A doubling of production isn't happening anytime soon. If anything, as ICE cars lose market share to EVs, they might become more expensive to produce due to diseconomies of scale (e.g., underutilization of factories, higher prices on car parts due to smaller orders, et cetera).

E. S-Curve

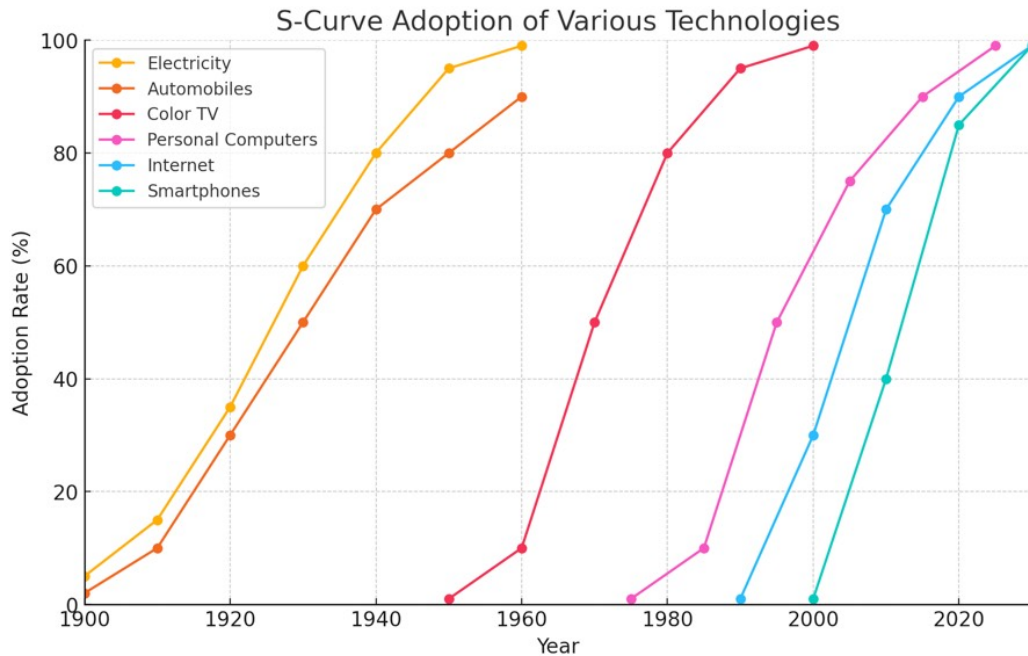


New technologies aren't adopted linearly but exponentially. When an emerging technology is significantly better, vastly more useful, and/or cheaper than an existing technology, it will be adopted following an S-curve. The graph below illustrates how technologies are being adopted faster and faster over time.

As you can see, the adoption of new technologies is slow at first. It starts with the innovators. These are the people who bought the first PC, early mobile phones, or EVs when they were expensive and had lots of bugs. Then they were joined by the early adopters, who are quick to adopt new technology once they can see the clear benefits. The early majority waits for technology to be useful and extensively tested before adopting it. They are more conservative but practical. The late majority is skeptical but ends up adopting technology when it becomes cheaper and/or when they are forced to adopt because everyone else is using it. Finally, the laggards, being resistant to change, are the last to adopt technology. For instance, they are the ones using flip phones while everyone else has switched to smartphones.

S-Curve acceleration

As the world has become more connected, the adoption of new technologies has accelerated, often because new technologies are built on top of existing ones.



In the US, it took twenty years for the internet to go from one percent to ninety percent adoption. The smartphone grew from one percent to eighty percent adoption in just ten years. And ChatGPT reached one hundred million users in two months, making it the fastest-growing consumer application in history.

By understanding S-curve adoption, we can better predict the speed at which new technologies will grow. This is another tool we can rely on to think more accurately and make better decisions.

* * *

Action step

Imagine a pond with one lily pad. Every day, the number of lily pads doubles. In thirty days, the pond will be completely covered.

Question: On which day will the pond be half-covered?

List three areas where you might be thinking linearly (examples: technology adoption, career progression, investing, skill-building, health, et cetera). Then for each, ask:

“If this were to follow an exponential curve, how would things look 2X, 4X, or 10X further down the line?”

Now, how likely is it to follow an exponential curve?

Think of what industry or technology will likely grow exponentially in the next five to ten years. Then reflect on what the implications for society and for you

personally could be.

* International Energy Agency (IEA). "Evolution of Solar PV Module Cost by Data Source, 1970–2020." Accessed May 21, 2025. <https://www.iea.org/data-and-statistics/charts/evolution-of-solar-pv-module-cost-by-data-source-1970-2020>.

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LOOKING AT TRENDS AND PREDICTIONS

Predicting the future is one of the hardest things to do. At best, we can identify trends and be directionally correct. But even if we turn out to be right, we might just have been lucky.

To improve our ability to predict some aspects of the future, we can look at past predictions that were wrong. Inaccurate predictions mean that the person who made them may have failed to understand how the world works and made poor assumptions. Or it could be that the probabilities were on their side, but reality turned out differently due to the unpredictable nature of the world. We want to understand what happened to identify what they may have gotten right and what they may have gotten wrong.

Let's go over a few examples of bad predictions and see what lessons can be learned.

A. Availability of resources in the future

For decades, we've heard that the world will run out of resources. However, for the most part, this has not happened, so far. Yet many people are as convinced today as people were decades ago that we will run out of resources soon and that the world as we know it will collapse. It is a possibility. How likely is it, though?

The Simon-Ehrlich wager

Julian Simon, an economist, believed in human ingenuity. He was convinced that, rather than lacking resources, we will develop more resources in the future despite population growth. On the other hand, Paul Ehrlich, a famous biologist, warned of resource shortages and environmental disasters due to overpopulation. In his book, *The Population Bomb*^{*}, published in 1968, Ehrlich argued that hundreds of millions of people could die from starvation if we don't control population growth.

In 1980, Simon and Ehrlich made a wager that included the following commodities: copper, corn, cotton, sulfur, and wheat^{*}. Simon believed that the price of each of these commodities would go down due to innovation and human progress. Conversely, Ehrlich believed that, due to population increase, resources would

become scarcer, leading to higher prices. When they looked ten years later, the price had gone down for all five commodities. Simon won the wager.

As we've seen before, one of the biggest mistakes people make when trying to predict the future is to imagine that it will be like the present. They underestimate the ability of human beings to adapt and find solutions to even the most complex problems. In other words, they lack imagination. At any point in the last hundred years, we could have extrapolated the situation ten or twenty years into the future and seen insurmountable problems on the horizon:

"Hundreds of millions of people are going to die from starvation."

"We're going to run out of oil in the next ten years."

"The internet will never be able to scale."

"We won't have enough energy to run AI supercomputers."

Now, things could go wrong, and humanity could suffer terrible consequences from climate change or be wiped out because of a nuclear war or another pandemic. We can't predict cataclysmic events. But here is the lesson: by understanding our biases and learning from our failed predictions, we can become a little better at predicting future trends.

This is why it's important to understand cycles and patterns, such as exponential growth, the cost decline curve, Wright's Law, or the S-curve. Our inability to integrate these mental models into our thinking process can lead us to make inaccurate predictions. To improve our ability to predict the future, we must understand human nature and ingenuity as well as the mechanisms through which innovations occur and how they scale.

B. Climate change and catastrophic events

While there is a consensus among scientists that climate change is real and that human activities are the primary cause^{*}, we can't blame people for being skeptical. There is a long history of terrible predictions and fearmongering in the media. For instance, here is a real quote from an article published in the UK newspaper, *The Guardian*, more than twenty years ago:

"A secret report, suppressed by US defense chiefs and obtained by *The Observer*, warns that major European cities will be sunk beneath rising seas as Britain is plunged into a "Siberian" climate by 2020. Nuclear conflict, mega-droughts, famine, and widespread rioting will erupt across the world."

In 1989, Noel Brown, director of the New York office of the UN Environment Program, warned that, "entire nations could be wiped off the face of the Earth by rising sea levels if the global warming trend is not reversed by the year 2000."^{*}

A 2008 ABC News segment, *Earth 2100*, portrayed a dystopian 2015 where climate change would cause New York City to flood and Miami would need to be evacuated due to hurricanes and sea level rise.^{*}

To try to find the truth and make better predictions, we must look at the data and the trends. For instance, when it comes to climate change, are temperatures rising? Are natural disasters becoming more severe and more frequent? Is the sea level rising? Is ice melting in the Arctic? Is it due to human activities?

The answer to all of these questions is a resounding “yes.”

The point is this. The fact that many catastrophic scenarios announced by some people haven’t happened isn’t evidence that there is no observable change in the climate. Highlighting failed predictions (often made by activists rather than real scientists) is an example of cherry-picking data to fit our preferred narrative. On a side note, predictive models are more advanced and more reliable than they were decades ago when many outlandish predictions were made.

To conclude, in the same way we fall for biases that make us vulnerable to cults, scams, or extreme ideologies, we also fall for biases that lead us to make poor predictions or to believe in predictions that aren’t based on reliable models. When trying to predict the future, let’s look at the actual data and use a variety of useful models. We won’t get it entirely right, but we might get it a little less wrong. Meanwhile, let’s be skeptical of people who cherry-pick data or share anecdotes or personal experiences to support their case.

* * *

Action step

Choose one of the failed predictions mentioned in this chapter or find one yourself (use a search like “failed tech predictions” or “inaccurate climate predictions”).

Example: “We’ll run out of oil in the next ten years.”

Answer the following:

- What exactly was the prediction?
- When was it made? Who made it?
- What assumptions did they rely on?
- What actually happened?
- What lesson can you draw from this case?

Think of one thing you believe will happen in the next ten to twenty years. Write it down.

Then answer the questions below:

- What trends am I basing this prediction on?

- What assumptions am I making about human behavior, innovation, or resource availability?
 - How might I be wrong? What could surprise me?
-

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* Simon, Julian L., and Paul Ehrlich. "The Bet." In *The Ultimate Resource 2*, by Julian L. Simon, 32–35. Princeton: Princeton University Press, 1996.

* Church, John A., et al. "Sea Level Change." In *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by T.F. Stocker et al., 1137–1216. Cambridge: Cambridge University Press, 2013.

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* ABC News. "Earth 2100." Aired June 2, 2008.

CONCLUSION

If you don't think for yourself, someone else will tell you how you should think. In this book, we've seen that critical thinking isn't optional.

In the first part, we saw that our desire to belong, be right, feel special, and seek congruence often leads us to abandon critical thinking. Wanting to be accepted by our peers, we make irrational decisions that work against us. Keen to be right, we refuse to challenge our beliefs and only seek to confirm what we already believe. Wishing to be special, we buy into various conspiracy theories that give us the illusion we know something that others don't. Striving to protect a certain image of ourselves, we fall for the commitment bias, the sunk cost fallacy, or we rationalize after the fact.

In the second part, we discussed how to separate facts from feelings. We went over the hierarchy of facts and explored how to assess the reliability of the information we consume so that we can sharpen our thinking. In addition, we learned how to differentiate between facts and assumptions.

In the third part, we identified how to look at data more objectively. We discussed how to avoid information overload. We explored the issue with confirmation bias. We analyzed why ideologies can be so appealing. We explained how statistics can be misleading, and we stressed the importance of thinking in terms of probabilities to improve our decision-making process.

In the fourth part, we investigated ways to protect ourselves against being manipulated. We warned against false experts and how they manipulate us by cherry-picking data, relying on anecdotes, using flawed analogies, oversimplifying things, or appealing to authority. We delved into conspiracy theories, analyzed the mechanisms behind them, and offered tools to assess their validity. Then, we explored the world of MLMs, why people fall for them, and how to protect ourselves against them. Finally, we dissected various scams and other dubious "opportunities."

In the fifth and final part, we introduced a variety of rules and frameworks to help us make better predictions and wiser decisions. We saw how we often fail to

understand exponential growth and how we tend to underestimate the speed of innovation.

If you've made it this far, you should have a better understanding of the main biases that prevent you from thinking for yourself. It's now up to you to refine your thinking. Remember, nobody is immune to manipulation. Manipulators will wait for you to lower your guard so that they can take advantage of you. They'll promise you quick results, offer a magic pill for all your problems, or try to pitch you an ideology or vision of the world that is wildly inaccurate. Understanding your biases and thinking for yourself is the antidote against such manipulation. Time is your most precious resource. By thinking more critically, you can avoid going down the wrong path, whether it's believing in far-fetched conspiracies, embracing extreme ideologies, or falling for scams.

Remember, if you stop thinking for yourself, someone will think for you.

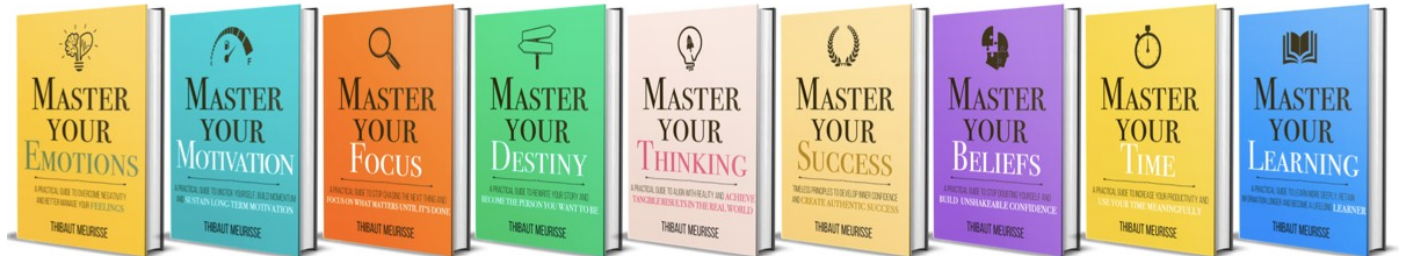
Always think for yourself.

Wish you all the best.

Thibaut

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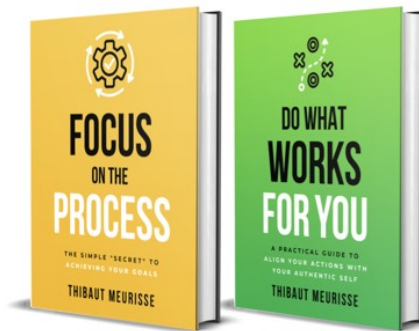
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Success is Inevitable: 17 Laws to Unlock Your Hidden Potential, Skyrocket Your Confidence and Get What You Want From Life

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